Numerical Analysis of an Industrial Scale Continuous Container Glass Annealing Furnace

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❖ Introduction

- ❑ Annealing is a type of heat treatment process used in glass manufacturing.
- ❑ Rapid cooling of glassware during the forming results in residual thermal stresses. Annealing is used to relieve these stresses.
- ❑ In high volume glass manufacturing annealing process is carried out in continuous furnaces.
- ❑ A precise temperature control of the process is essential for the quality.

Forming process

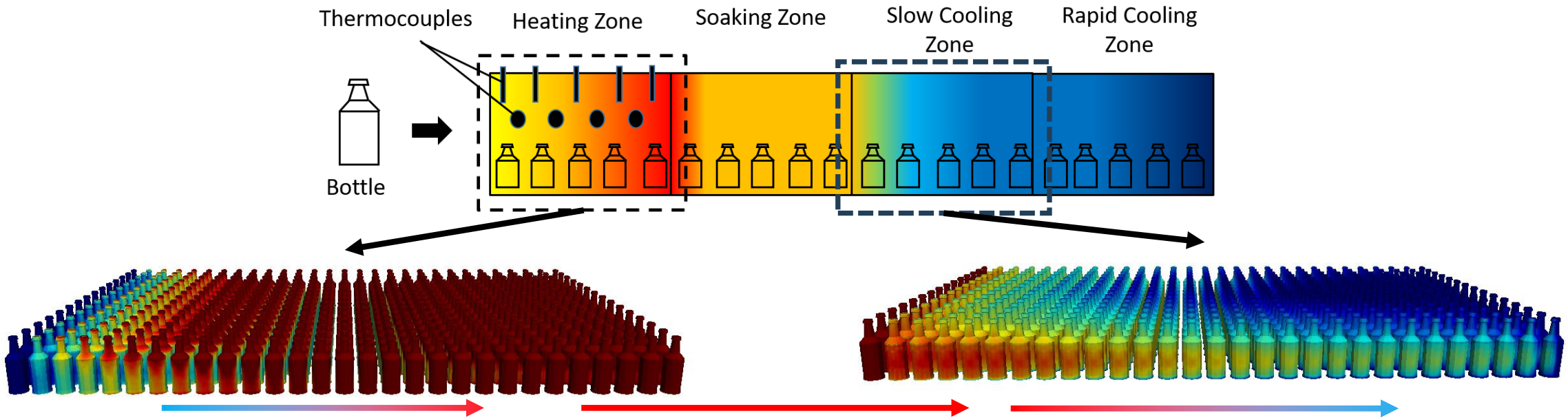
Source: https://www.haberturk.com/anadolu-cam-in-rusya-daki-fabrikasi-uretime-yeniden-basliyor-2509360-ekonomi

Annealing Furnace Inlet

Source: https://stdglass.az/ru/

❖ Continuous Glass Annealing Furnace

- ❑ Continuous Annealing Furnaces have different zones.
- ❑ **Purpose of the heating zone :** to heat up the glass to the temp. at which the internal stresses can relieve. (Annealing temp)
- ❑ **Purpose of the soaking zone :** to even out the temperature non-uniformities in glass formed in the heating zone.
- ❑ **Purpose of the cooling zone :** a glass annealing furnace gradually cools hot glass, relieving stress, stabilizing dimensions, enhancing strength, ensuring consistency, and preventing thermal shock. This process produces high-quality, durable glass products.



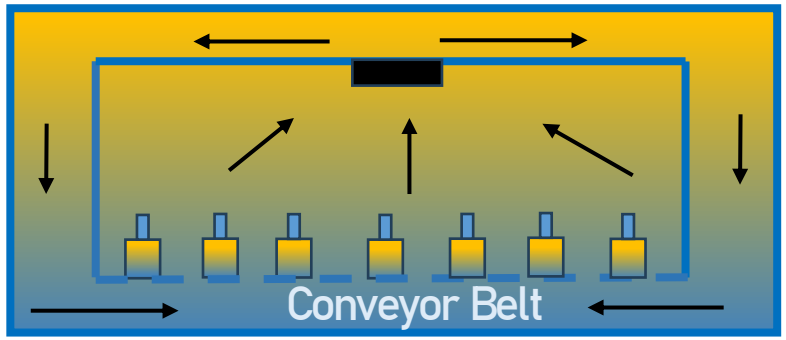
Schematic annealing furnace

❖ Heat Transfer Mechanism inside the furnace

- ❑ Heating is provided by the tube burners

- ❑ **Furnace has a double layer structure :** A circulation fan mounted on the top wall circulates flow in the furnace.

Cross-section view



A typical annealing furnace viewed from inlet

- ❑ 3 modes of heat transfer
  - Conduction - > within the glass products
  - Convection -> within the furnace that is driven by circulation fans
  - Radiation -> from all surface
    - between furnace walls
    - between glass products
    - between glass and furnace walls

❖ Gaps in Literature

- Even though relatively more complex models exist for steel annealing, glass annealing has lagged behind in terms of modeling complexity.
- Most Works apply Small Operational Loads, Simplified Geometries or utilize only one or two heat transfer modes per simulation

- To date, no study regarding a CFD Model **specifically** for a Glass Annealing Furnace was found that **simultaneously** includes:
  - A Real Operational Load
  - Actual Glassware and Furnace Geometries
  - A conjugate solver which solves for conduction, convection and radiation heat transfer modes

❖ Objectives of the Research

- ❑ Develop a physically accurate model of an industrial-scale contionous glass annealing furnace
- ❑ Resolve, Conduction, Convection and Radiation, with Real Furnace and Glassware Geometry and with Real Operational Load
- ❑ Assess the effects of different burner configurations on Zone Temperatures, Product quality and energy efficiency
- ❑ Provide Industry Applicable insights for optimizing energy efficiency and operational quality

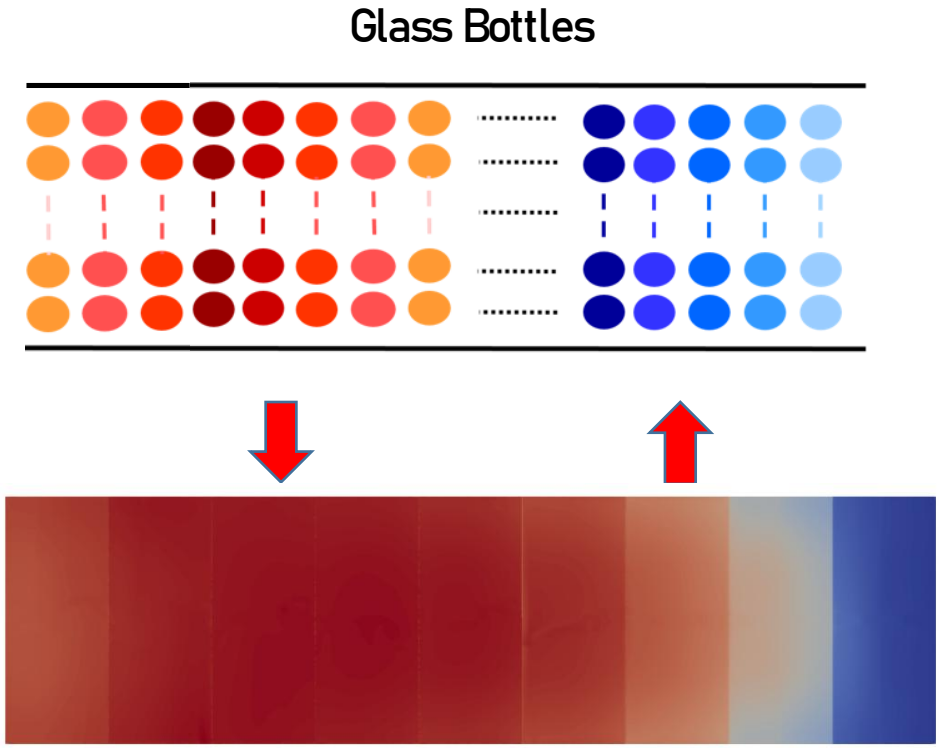
❖ Numerical Methodology (1) : Physics and Models

❑ **Conjugate heat transfer problem :**

- Solid Domain : Glassware inside the furnace (Transient)
- Air Domain : Atmosphere inside the furnace (Steady)

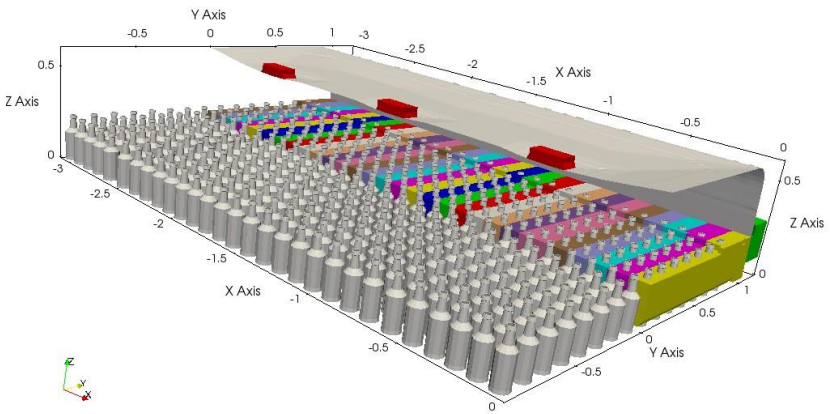
	Solid Domain	Air Domain
Heat transfer physics	Unsteady Conduction	Steady Convection
Governing Equations	Transient Heat Conduction	Compressible RANS with k-ω turb. model
Frame of reference	Moving	Stationary
Thermal BC's	Radiation and Convection	Radiation and Convection
Wall BC's	No slip	No slip
Additional Models	-	Porosity based modeling for glass rows
Solver	OpenFoam with in-house S2S Radiation Model	OpenFoam with in-house S2S Radiation Model

❖ Numerical Methodology (2) : Steady Solver for air domain



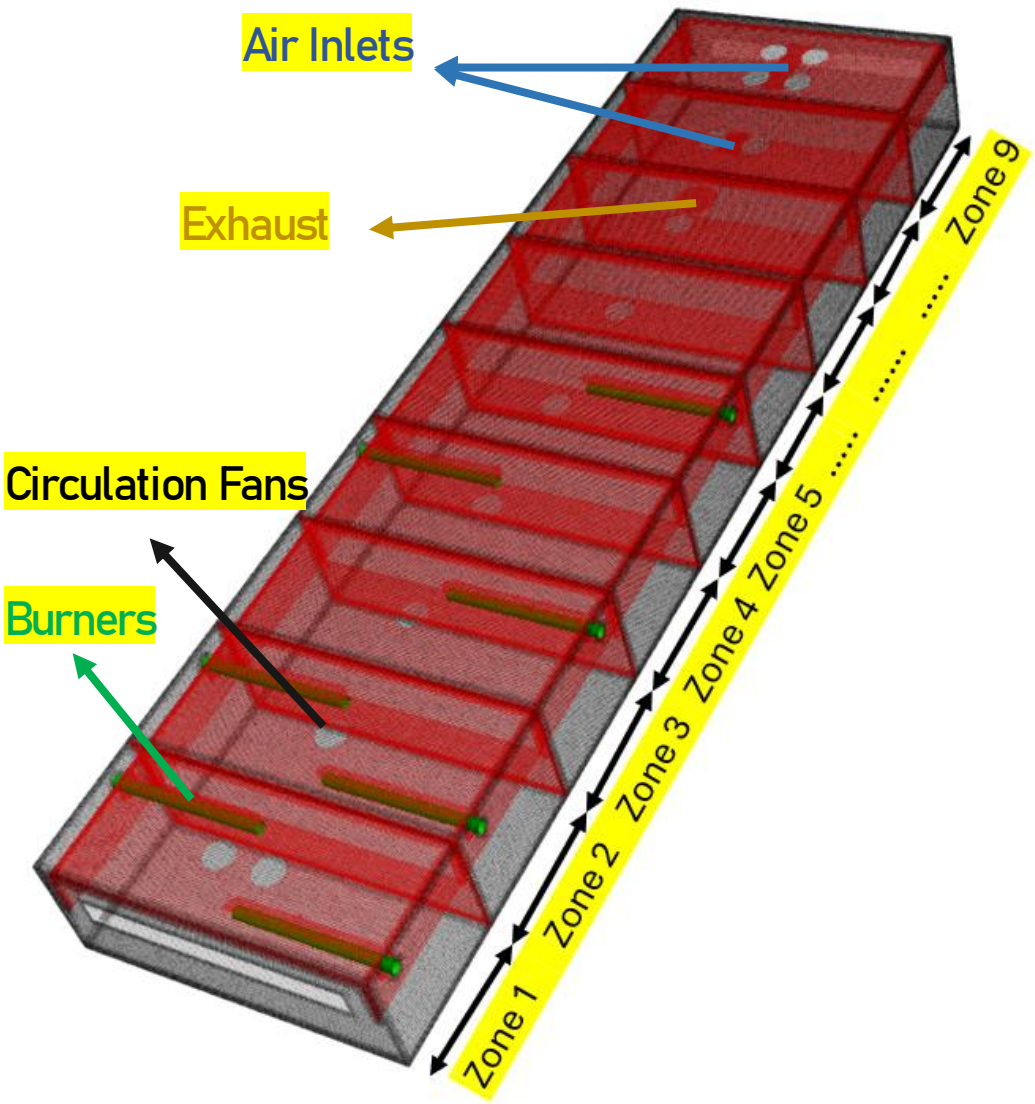
- ❑ Computational domain includes
  - Air
  - Bottle walls for radiation calculation

- ❑ Convection btw. air and glass is modelled via porosity based approach.
  - Darcy-Forchheimer coefficients are derived.

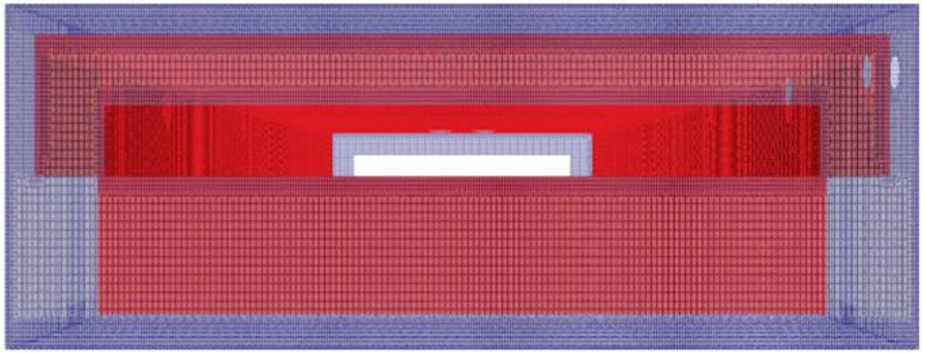


- ❑ Heat transfer coefficient of single bottle is applied to every other bottles.
- ❑ Backward Monte Carlo Ray Tracing method is utilized for radiation.

❖ Numerical Methodology (3) : Steady Solver for Air Domain – Convective Solver



- ❑ OpenFOAM's buoyantSimpleFOAM is used as a solver which applies the SIMPLE Algorithm for solving RANS equations
- ❑ The Furnace Walls has a Double Layer Structure, in which Inner Furnace Wall has modeled as a Baffle.



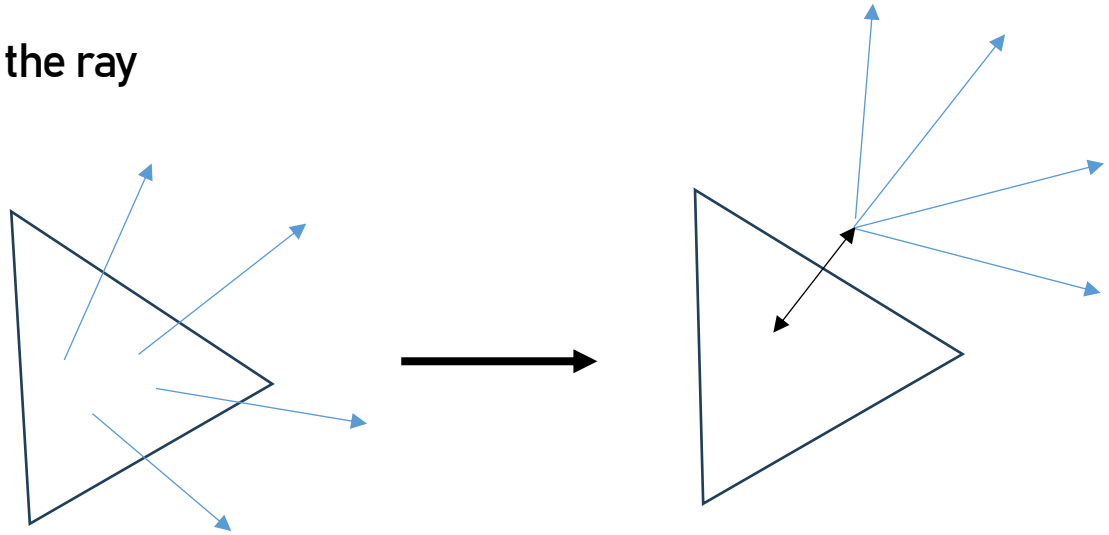
- ❑ The Air Inlets are modeled using pressure directed inlet velocity BC with the combination of Darcy-Forchheimer Resistance.
- ❑ Exhaust is defined with an inletOutlet BC.
- ❑ Circulation Fans are modeled using meanVelocityForce BC, where mean velocity vector $\vec{U}_z = 20\ \mathrm{m/s}$ is set for each Fan

❖ Numerical Methodology (4) : Steady Solver for Air Domain – Radiative Solver

- ❑ S2S Radiative Solver Utilizes Backwards MCRT
- ❑ Step by Step:
  - ➢ Energy which will be emitted is divided into payloads for each ray : $Q_i^{\text{emit}} = \varepsilon \sigma A_i T_i^4, \longrightarrow E = \frac{Q_i^{\text{emit}}}{N_{\text{ray}}}.$
  - ➢ Ray Direction is Randomized based on random seeds and cosine-weighted sampling over the hemisphere
  - ➢ Ray origin is determined using an edge-length weighted centroid method: $\vec{x}_{\text{origin}} = \frac{a\vec{v}_0 + b\vec{v}_1 + c\vec{v}_2}{a+b+c} + \delta\vec{d},$
  - ➢ After a hit is detected by the closest hit shader, the shader computes the corrected absorbed energy as:

$$E_{\text{abs}} = E\left(\frac{T_{\text{hit}}}{T_{\text{emit}}}\right)^4$$

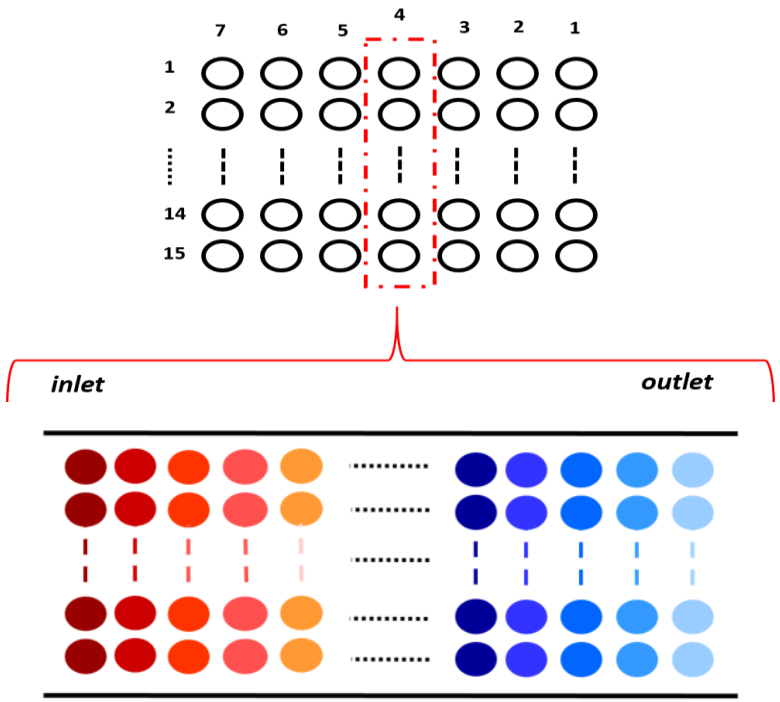
  - ➢ The absorbed energy then saved into the primitive which emitted the ray hence Backwards MCRT



Ray Origin Using Barycentric Method

Ray Origin Using Centroid Method

❖ Numerical Methodology (5) : Transient Solver for solid domain



- ❑ **Computational domain**
  - → consists of 7 rows of bottles instead of all of the rows.
  - → each row has 40 bottles
- ❑ Temperatures of glasses in the 4th row are stored to be passed to the steady solver.
- ❑ Transient conduction solver with radiative and convective bc's at the walls.
- ❑ Surface-to-surface radiative heat transfer
  - Backward Monte Carlo Ray Tracing model is utilized.
  - Surfaces are black and diffuse.

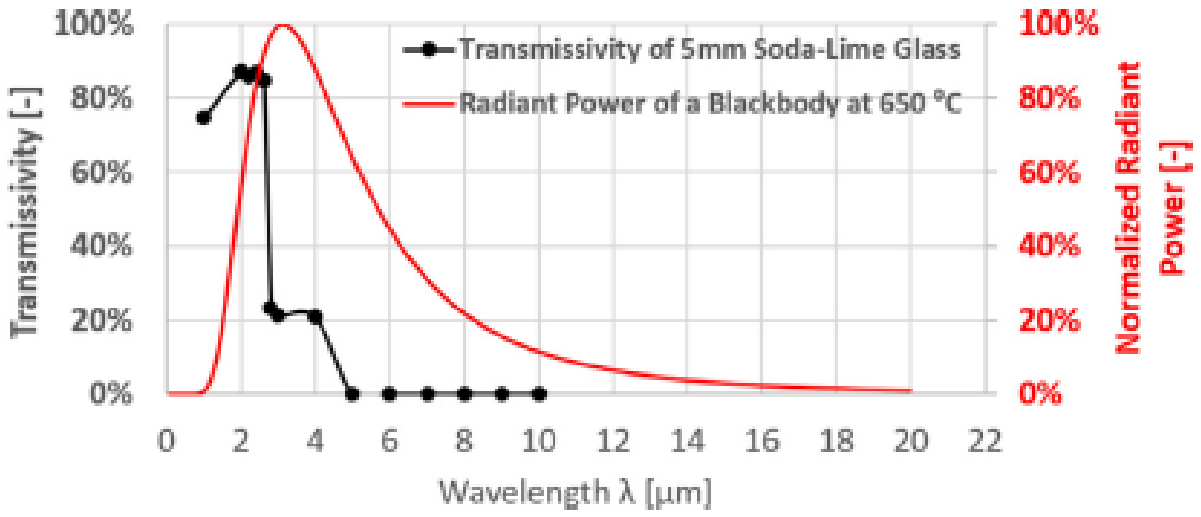
[1] Gönenç Can Altun, Altuğ Melik Başol, *Numerical analysis of the radiant heating effectiveness of a continuous glass annealing furnace*, App. Therm. Eng., 204, 2022

❖ Numerical Methodology (6)

- According to Yıldız et al., (2020), 5 mm thick Soda-lime glass act as a black body for wavelengths between 2.8µm and 5µm absorbing %80 of the emitted radiation

- %82 of the total radiation emitted from furnace walls are above 2.8µm in wavelength, resulting %18 overprediction when modelling the glass as black

- For simplification all of the surfaces are modeled as **black**

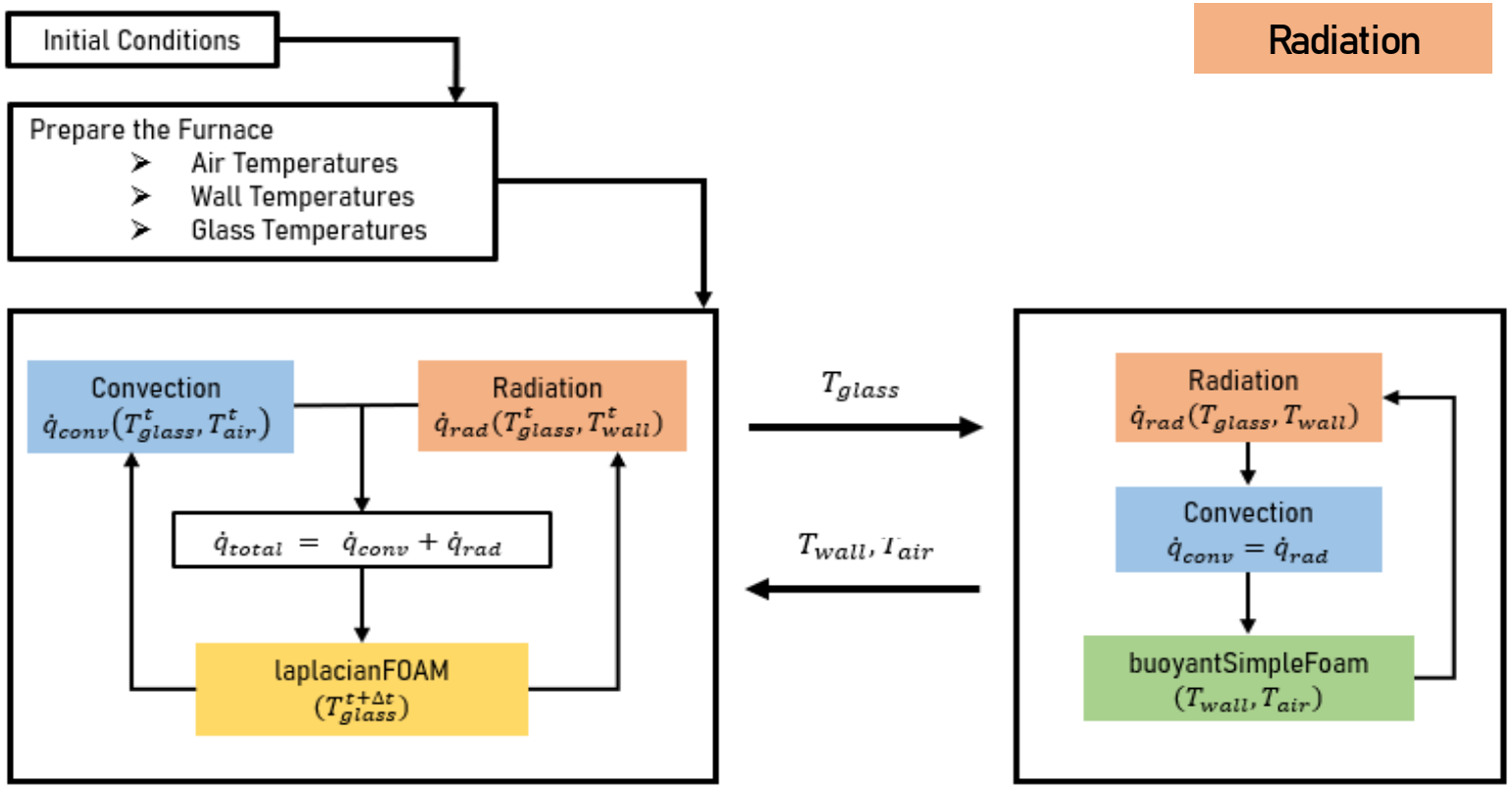


Spectral distribution of radiant power and transmissivity of 5 mm thick soda-lime glass vs. wavelength. (Ersin Yıldız, Altuğ Melik Başol, M. Pınar Mengüç, Segregated modeling of continuous heat treatment furnaces,Journal of Quantitative Spectroscopy & Radiative Transfer,2020)

❖ Numerical Methodology (7)

❑ Two separate solvers used for the solid and air domains.
- ➢ Transient solver for solid domain
- ➢ Steady solver for air domain

Mode of heat transfer	Processing Unit
Conduction	CPU
Convection	CPU
Radiation	GPU



Transient Solver (for Solid Domain)

Solver Coupling

Steady-State Solver (for Air Domain)

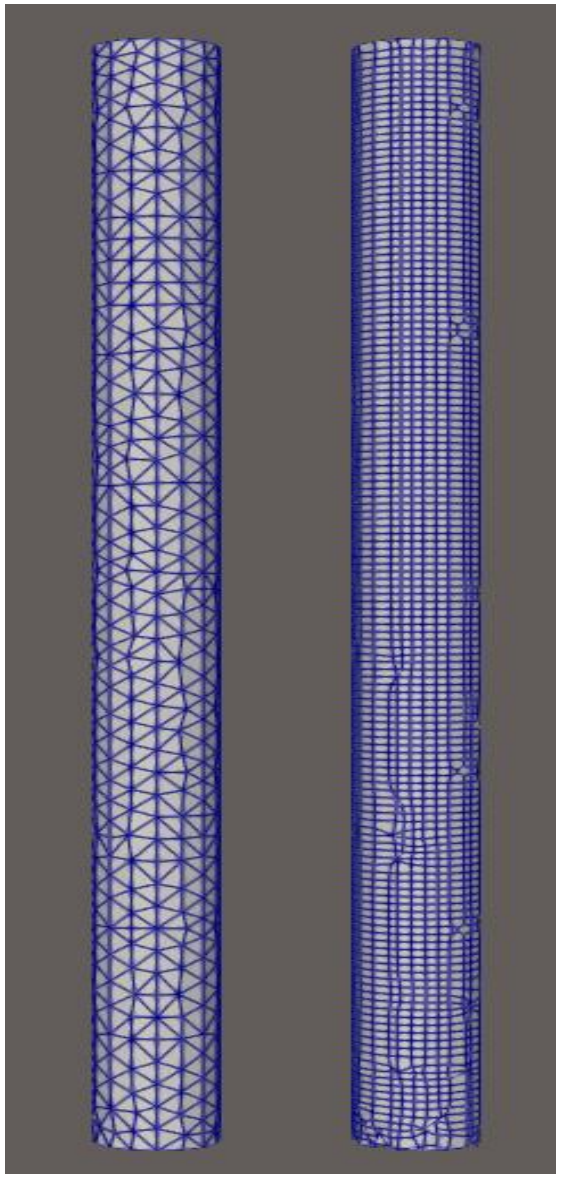
❖ Numerical Methodology (8)

❑ Two separate solvers are coupled utilizing Mapping of the Heat Fluxes and Temperatures

- ➢ Net Energy Flux is calculated as : $\dot{q}_{\text{net}} = \dot{q}_{\text{rad}} + \dot{q}_{\text{conv}}$
- ➢ Which From S2S mapped onto OpenFOAM
- ➢ externalWallHeatFluxTemperature is utilized to calculate Wall temperatures based on Heat Fluxes

❑To map the fluxes and the temperatures between the solvers, interpolations are made

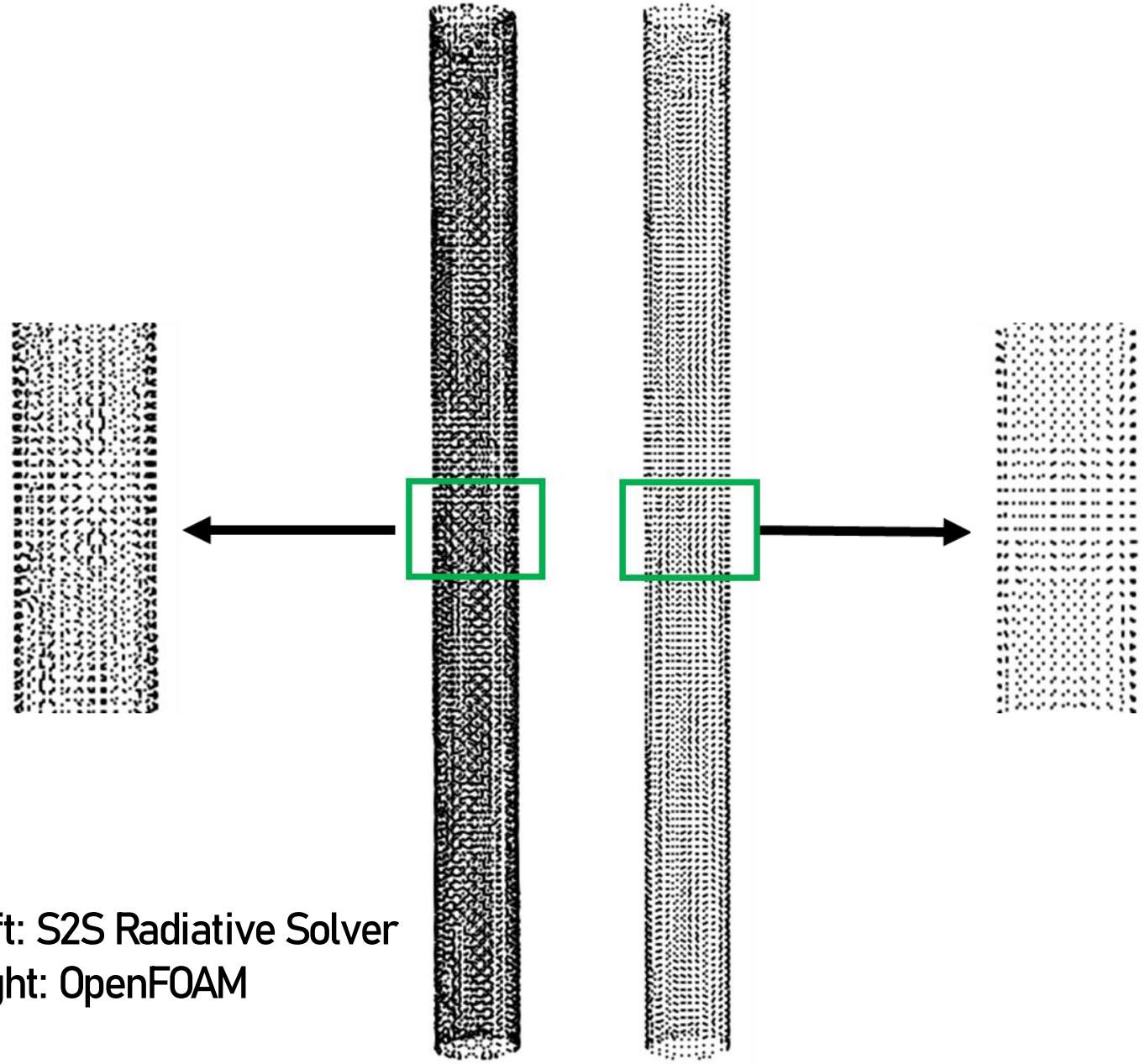
- ➢ Specifically: Utilizing SciPy griddata function where the method is set to closest



Left: S2S Radiative Solver
Right: OpenFOAM

❖ Numerical Methodology (9)

- ❑ Each Point Represents the Cell Centers of Each Primitive, and the corresponding Temperature Data
- ❑ The Interpolator utilizes the Cell Centers of both S2S and OpenFOAM meshes to find the closest cells
- ❑ Step by Step:
  - ➢ Radiative Heat Fluxes are mapped onto OpenFOAM mesh
  - ➢ Convective Flux is calculated and added to find the net heat flux by OpenFOAM, which then calculates the Temperatures
  - ➢ Temperature data is mapped back onto S2S Solvers Mesh

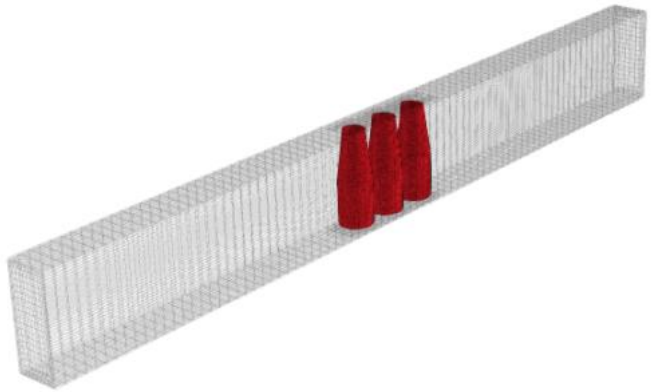


Left: S2S Radiative Solver
Right: OpenFOAM

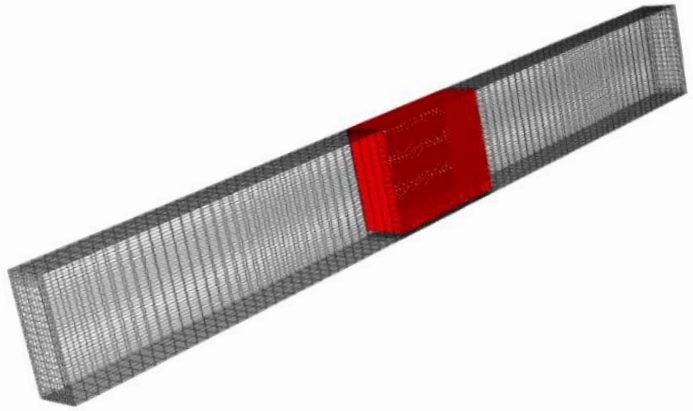
❖ Darcy-Forchheimer Porous Zone Modelling (1)

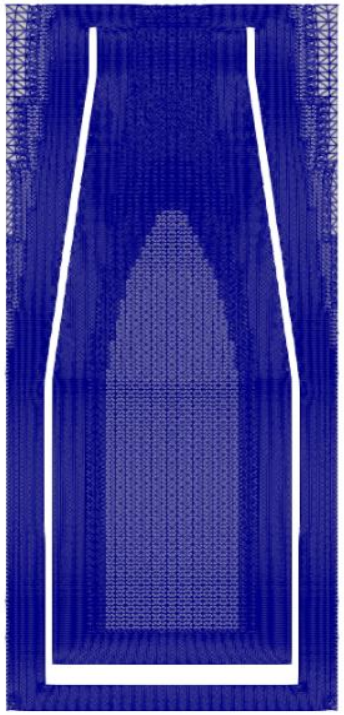
- ❑ The flow resistance caused by the glassware is modeled as a Darcy-Forchheimer Porous Zone
- ❑ Darcy-Forchheimer coefficients determined by simulations with 4 different Inlet Velocity Boundary Conditions, where simplified geometry is utilized
- ❑ The Tube's Walls are set as Slip BC
- ❑ The simulations repeated with Porous Zone and compared Pressure Drops vs Actual Bottles Results

Simulation with Actual Bottles



Simulation with Porous Zone





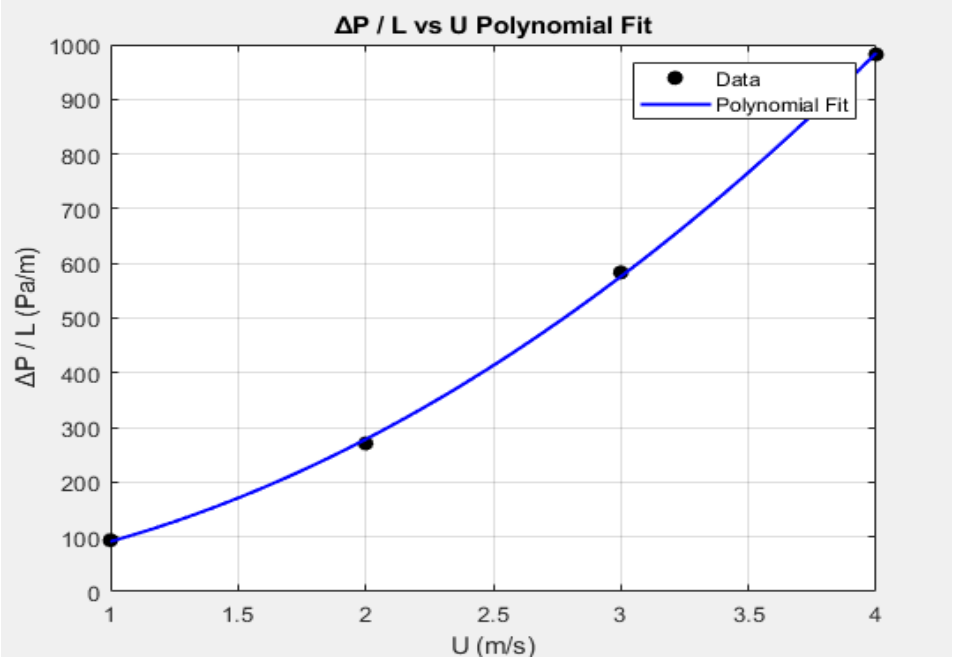
Mesh of the Bottle Zone

❖ Darcy-Forchheimer Porous Zone Modelling (2)

- According to the OpenFOAM's Darcy-Forchheimer model, if the velocity dependent pressure drop is known the following empirical relation can be utilized

$$\frac{\Delta p}{\Delta x} = Au + Bu^2 \xrightarrow{\text{Where}} A = \mu \mathbf{D} \quad \text{and} \quad B = \frac{1}{2}\rho \mathbf{F}$$

Following fit is utilized for the Darcy and Forchheimer coefficients (Using MatLAB's PolyFit function)

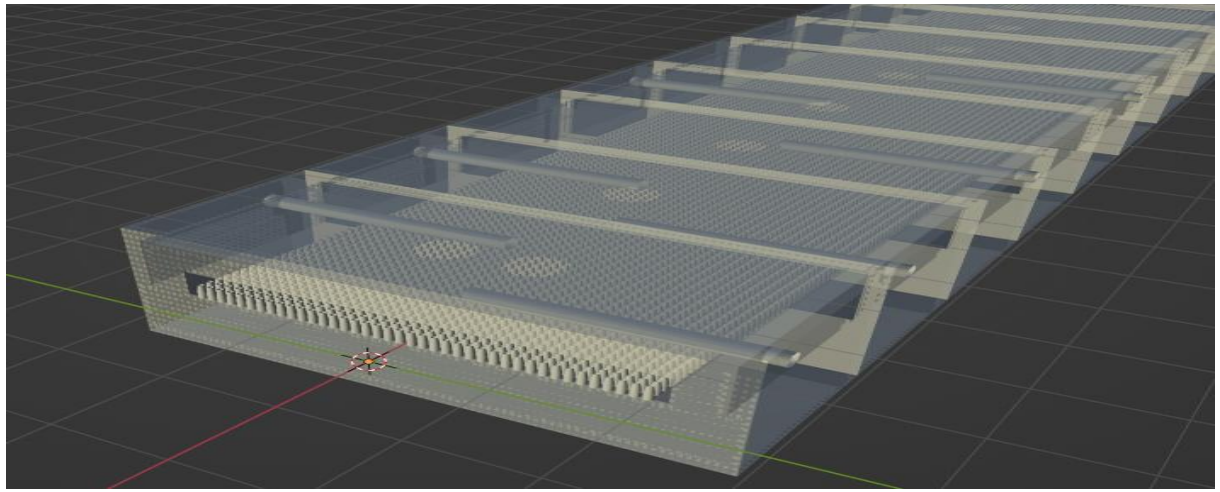


3 Bottles Polynomial Fit

- Following the Fit, the Forchheimer Coefficient is found as 93.6
- Darcy Coefficients are found as 0 as expected from a turbulent dominant flow.
- After the simulations validation simulation is made, initially 93.6 failed for agreement
- The Forchheimer coefficient has gradually increased to 123 which had agreement with the 3 Bottle Case, having 73.1 (With Glass objects) vs 73.6 (with Porous Zone)

❖ Computational Furnace Domain

- ❑ Annealing furnace is composed of 9 Zones : 5 Heating Zones and 4 Cooling Zones
- ❑ Heating zones are equipped with tube burners.
- ❑ Each zone has circulation fans.
- ❑ Cooling zones are equipped with air inlet openings and there is an opening for exhaust upstream of the cooling zones.
- ❑ 203 rows of bottles inside the furnace each → 8120 bottles in total



Dimensions of the Furnace

Length of the Furnace (L)	20.25 m
Width of Section (W)	5.2 m
Height of Section (H)	2 m
Distance between rows(D)	0.02 m
Length of one Zone (Lz)	2.25 m

Operating and Boundary of the furnace

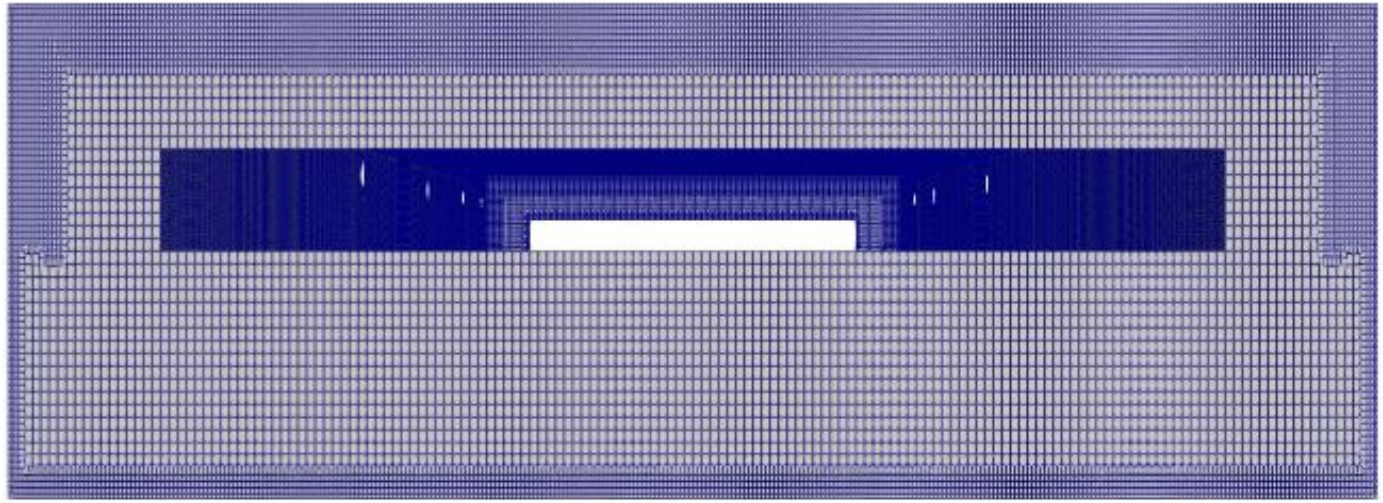
Glassware temp. at furnace inlet	477°C (uniform)
Active Burner tube wall temperature	677°C
Passive Burner tube wall temperature	527 °C
Radiative Properties of Furnace Walls	black
Radiative Properties of Glass	black
Conveyor Speed	60 cm/min
Cooling Air Inlet Mass Flow Rate	2.6 kg/s
Cooling Air Temperature	27°C

❖ Steady solver utilizes two different sets of Mesh.

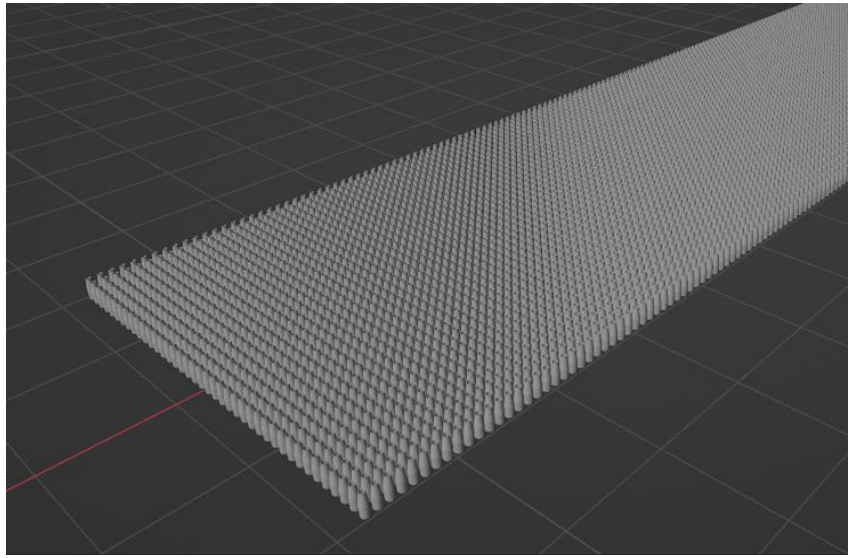
- **Air Domain Volume Mesh :** 4 Million Hexa Cells

- **Surface Mesh for Radiative Solver (S2S)**
  - Outer Furnace: 0.6 million Cells
  - Inner Furnace: 0.4 million Cells
  - Bottles: 10.2 Million Cells
  - Tube Burners: 57 Thousand Cells

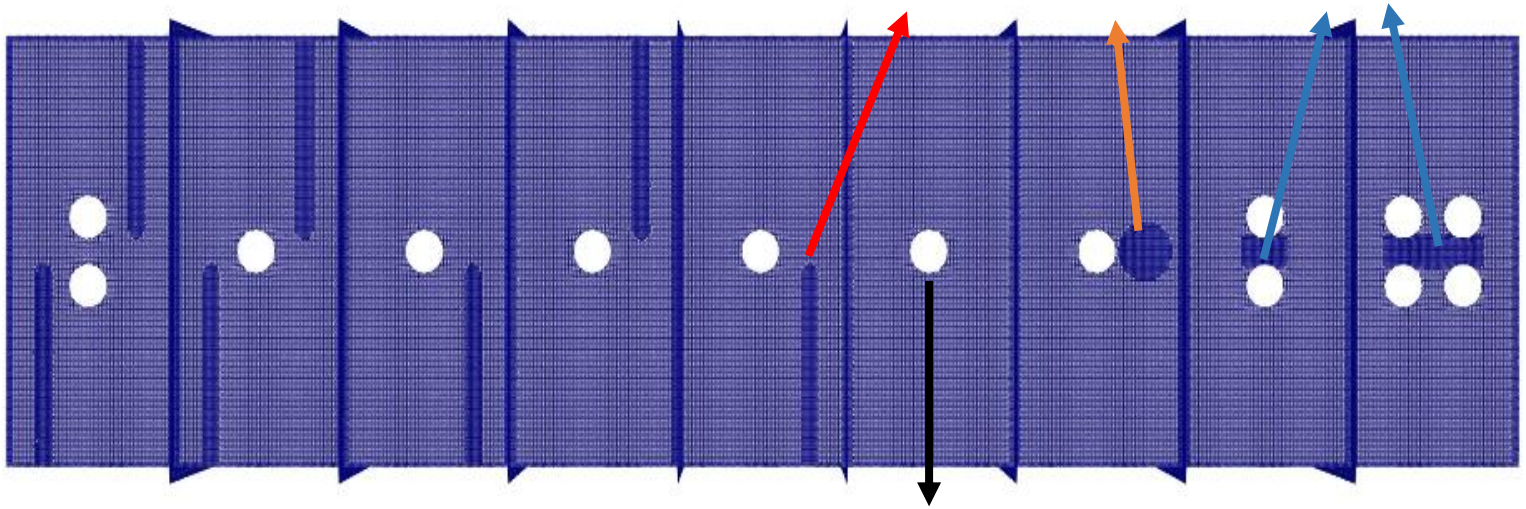
Outer Furnace Wall from the Inlet



Glass Bottles

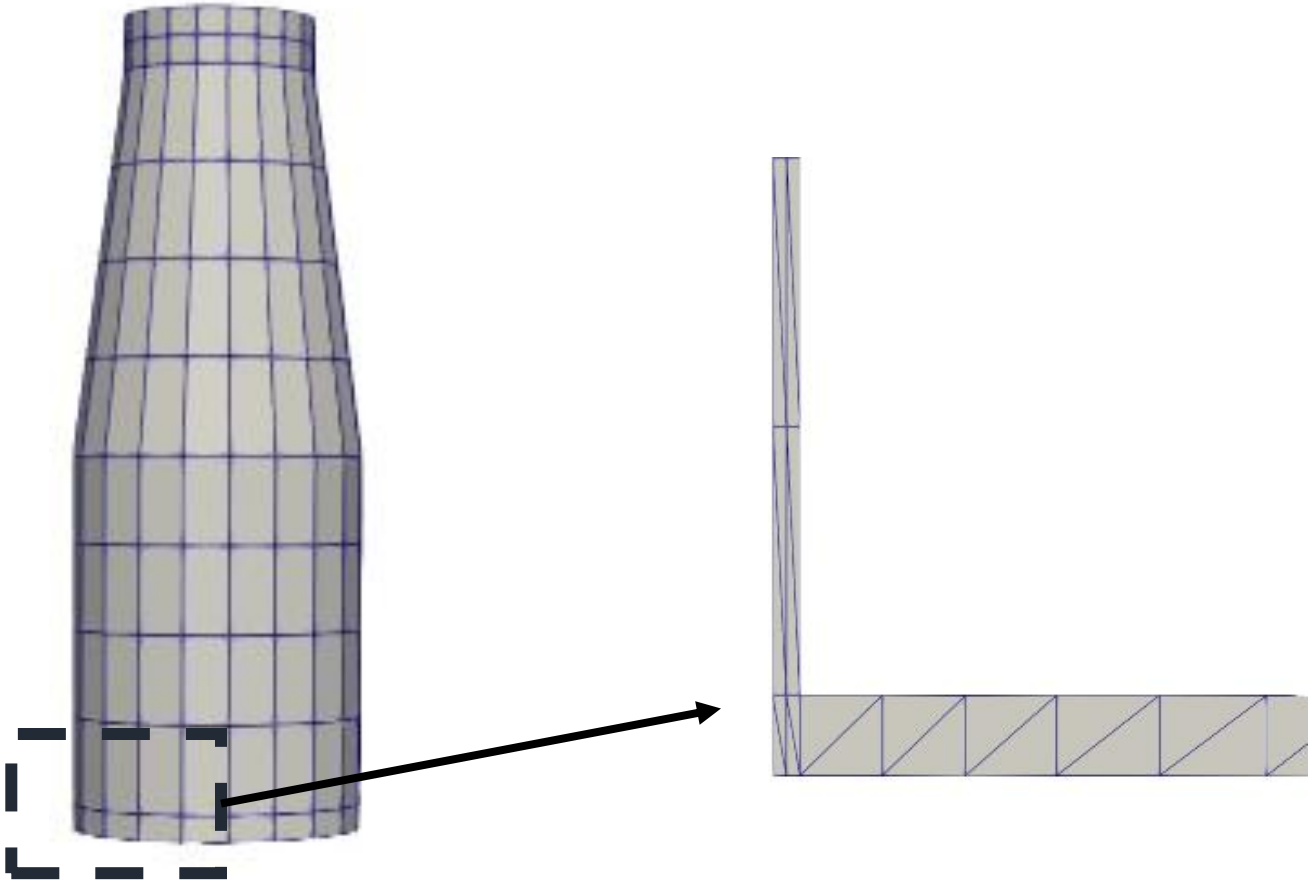


Inner Furnace Wall and the boundaries of the burners, exhaust and air inlets



❖ Transient solver mesh only includes 1 row of Glass bottles

- A relatively coarse volume mesh with 505 cells for a single bottle.
- Surface Mesh of a single bottle is 1260 Cells
- Each row of bottles includes 1260 x 40 surface mesh cells.

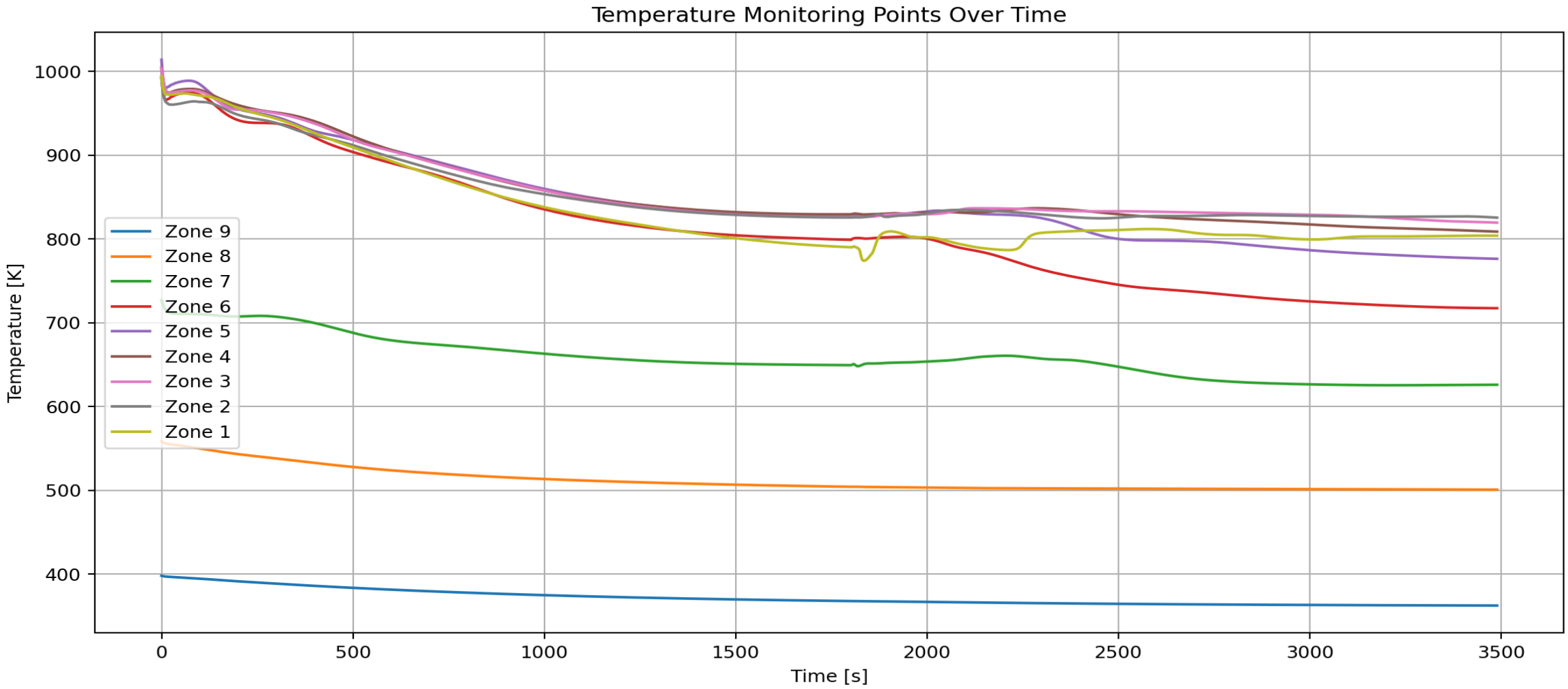


❖ Computational Cost

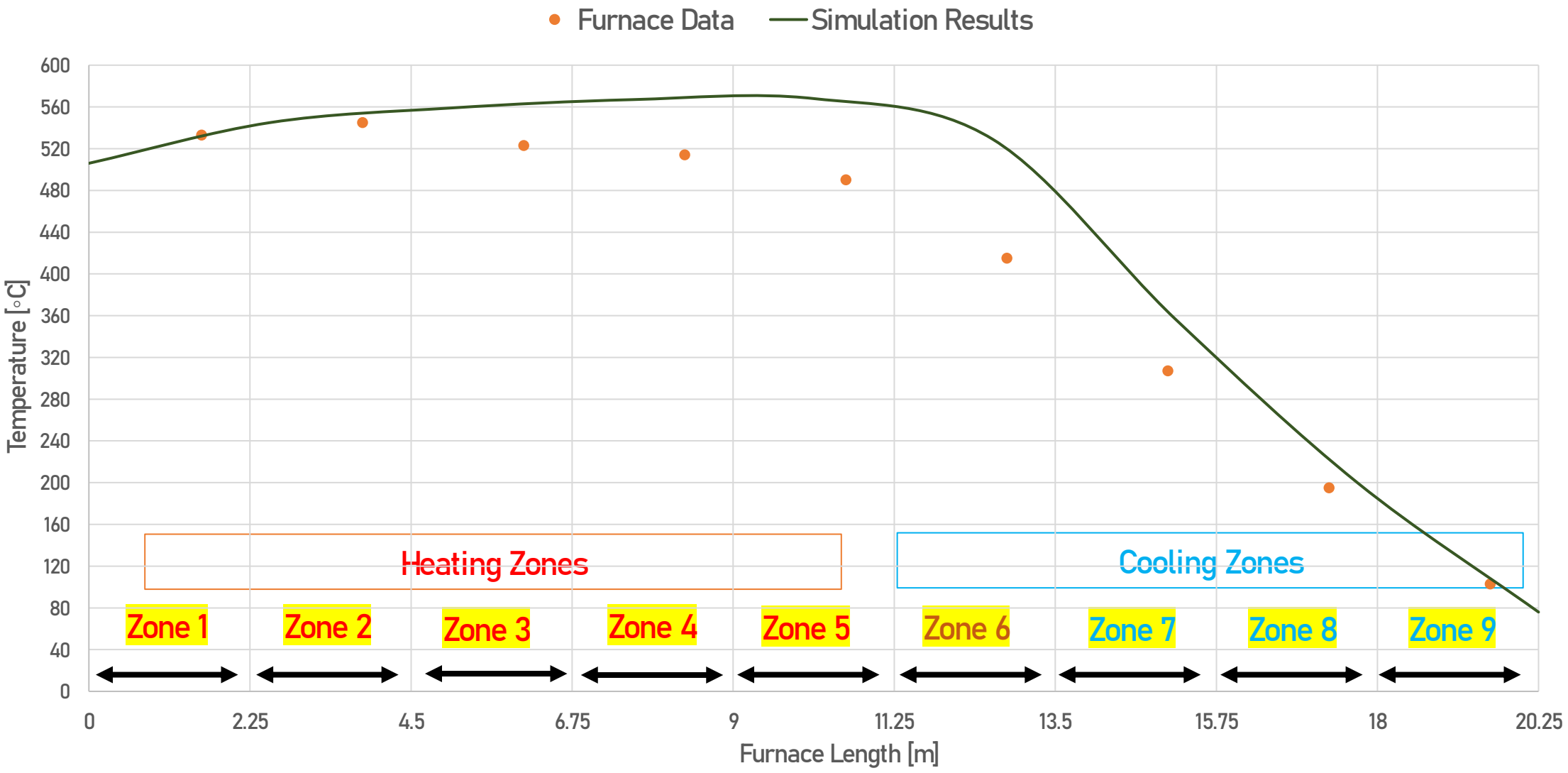
- ❑ The simulation is held on a **Regular PC**, (32 GB Ram, 8 Core CPU, NVIDIA Titan X GPU)
- ❑ The **Steady-State convective solver** operates on 8 different CPU cores
- ❑ The **Steady-State S2S radiation solver** emits the rays in parallel utilizing Nvidia Titan X GPU
- ❑ The Steady-State S2S radiation solver is called **every 10 Convective Iterations**

Solver	Time	Total Time
Steady Solver	~3 min (10 Convective 1 Radiative)	~17.5 hours (3500 Iterations)
Transient Solver	~1 min 10 sec (1 row of Glassware)	~4 hours (203 Rows of Glassware)

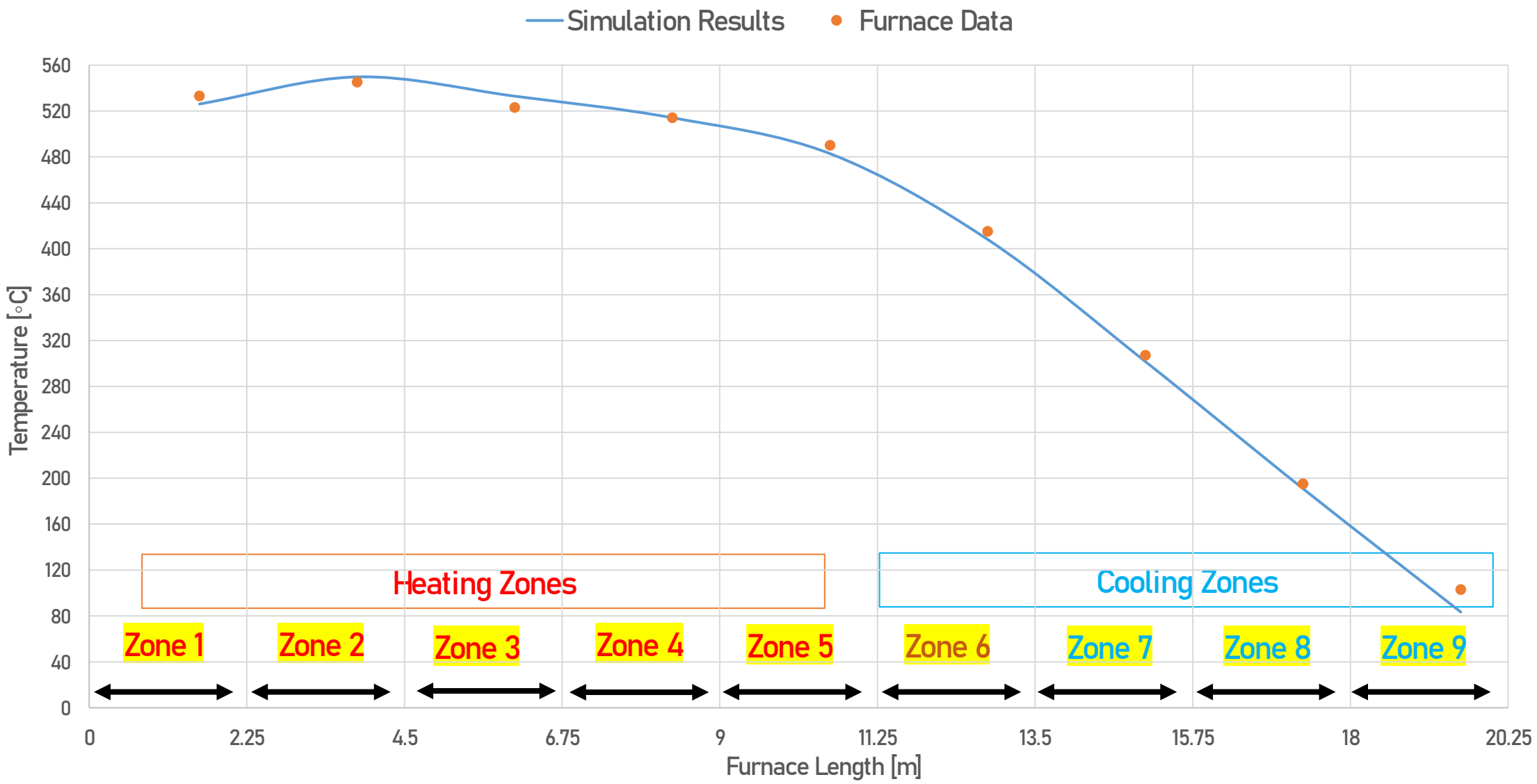
❖ Zone temperatures are monitored for convergence.



❖ Model Validation : Without Leakage from Glassware Inlet

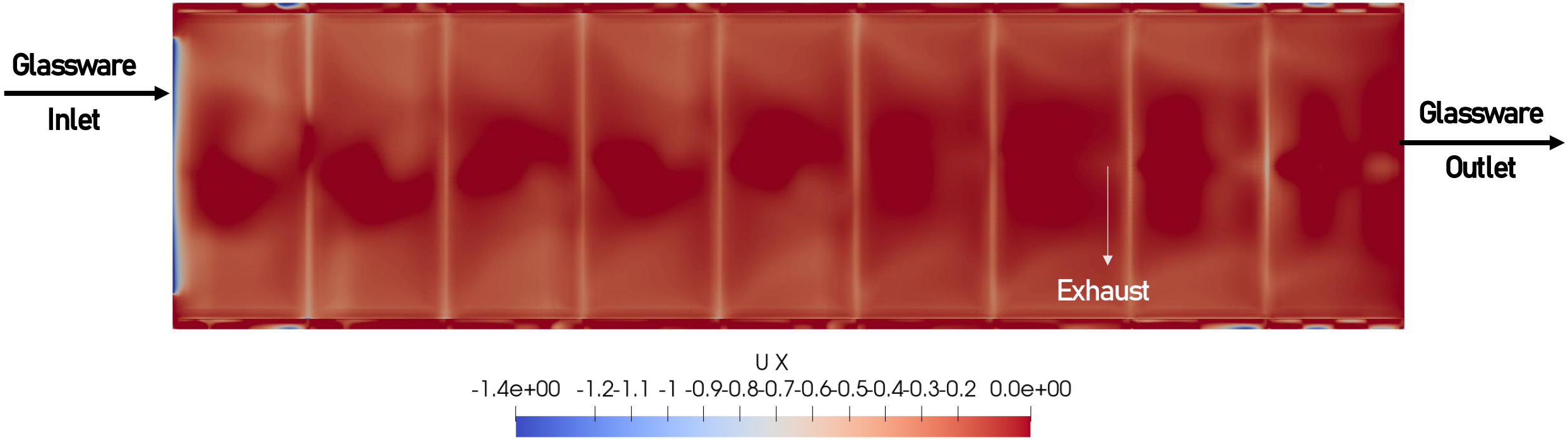


❖ Model Validation : With Leakage from Glassware Inlet



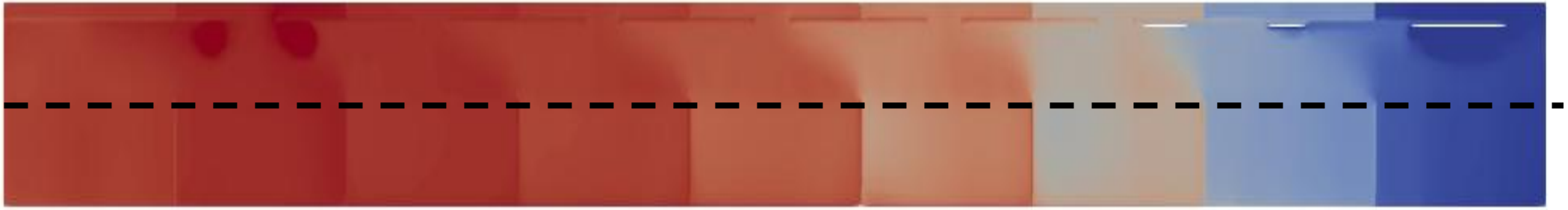
❖ Zone Air Velocity –x Directions

z-axis cut (Origin 0.1m)



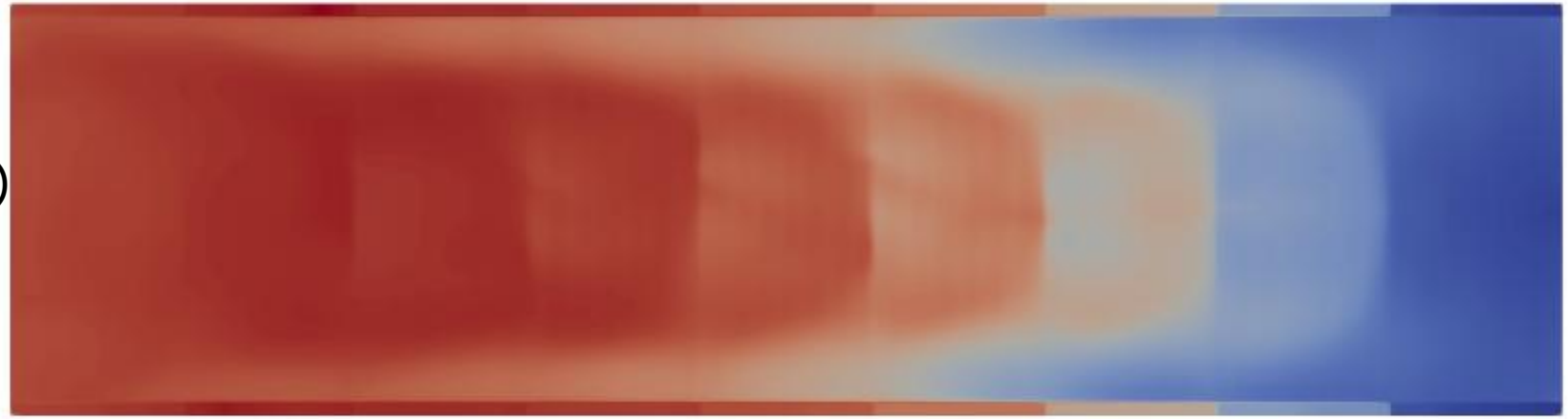
❖ Zone Air Temperature Distribution

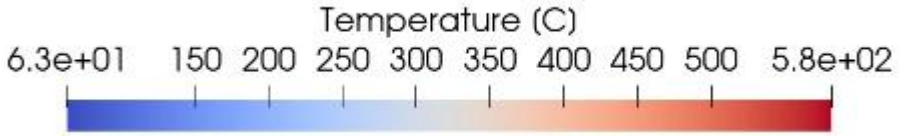
Furnace Midpoint Cut



z-axis cut location

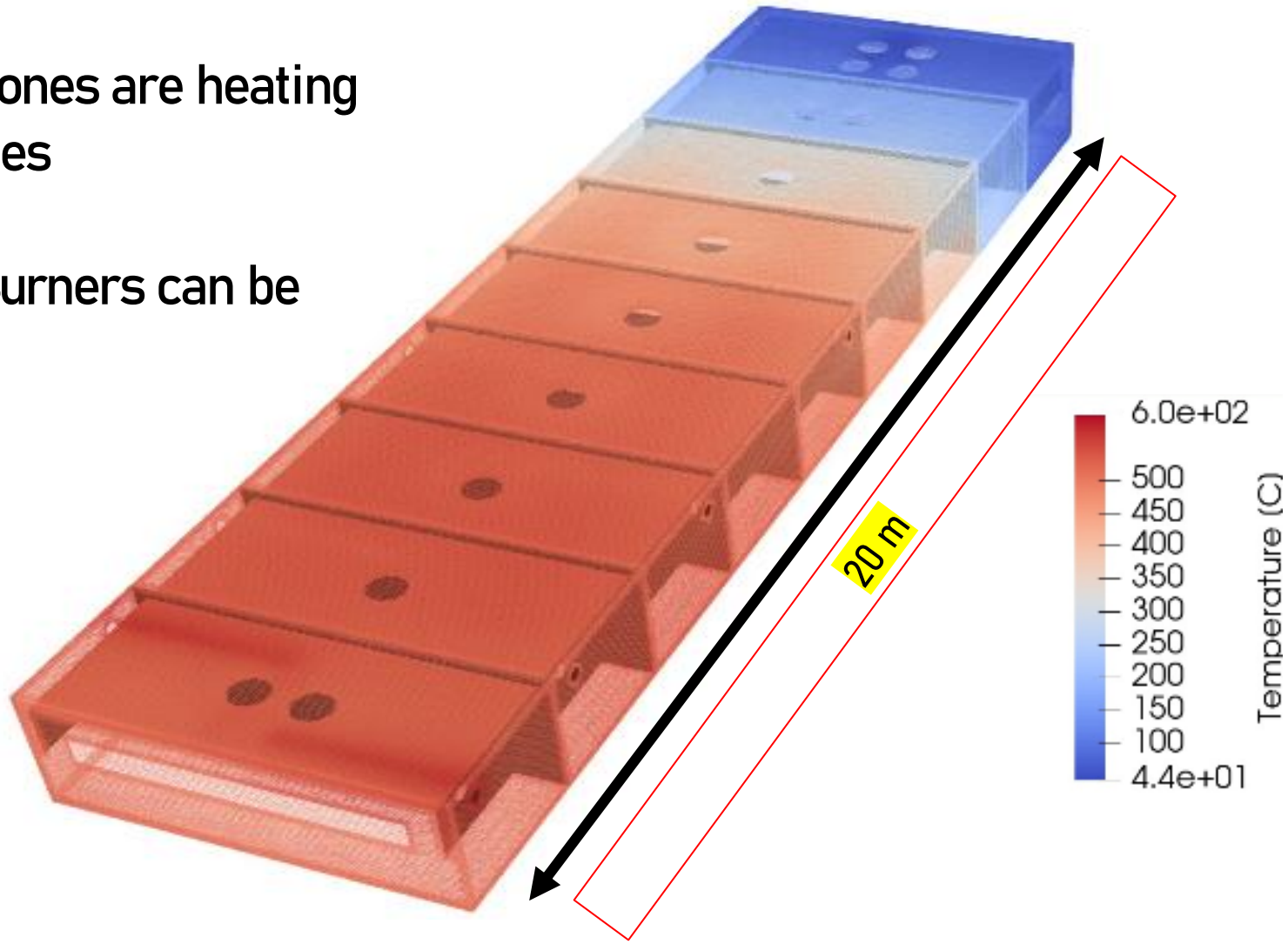
z-axis cut (Origin 0.1 m)





❖ Furnace wall temperatures and the radiative effect of tube burners on the inner furnace wall

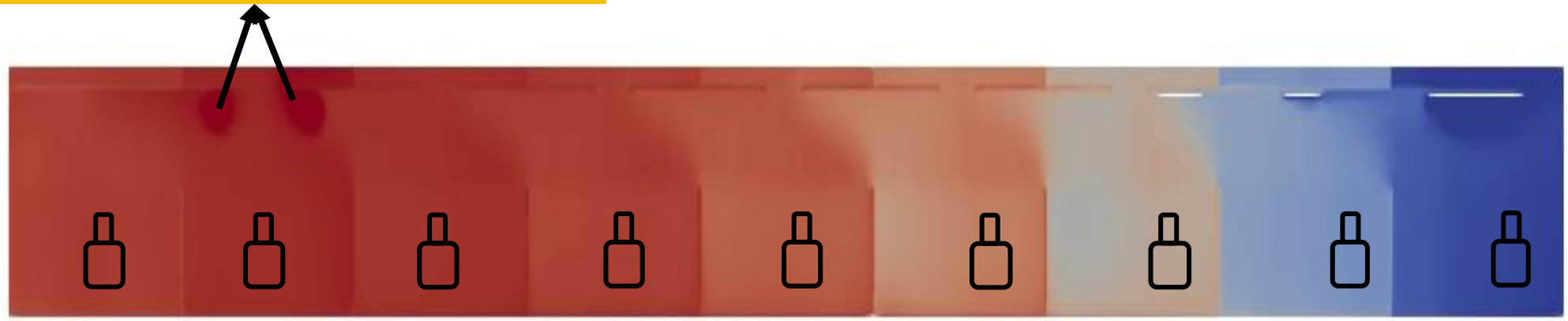
- 9 zones. From left to right first 5 Zones are heating zones and the last 4 is cooling zones
- The Radiative effects of the Tube Burners can be seen on the inner furnace walls



❖ Parametric Study

Case 1 : Active Tube Burners in Zone 1

Case 2 : Active Tube Burners in Zone 2



❖ Burner Boundary Conditions

- ❑ Burner Walls are prescribed with Fixed Wall temperatures, differentiating for Active and Passive burners
- ❑ Inlet Boundary Conditions are utilized for modelling the convective effect
- ❑ Following BCs are utilized for S2S Radiative and Convective Solvers

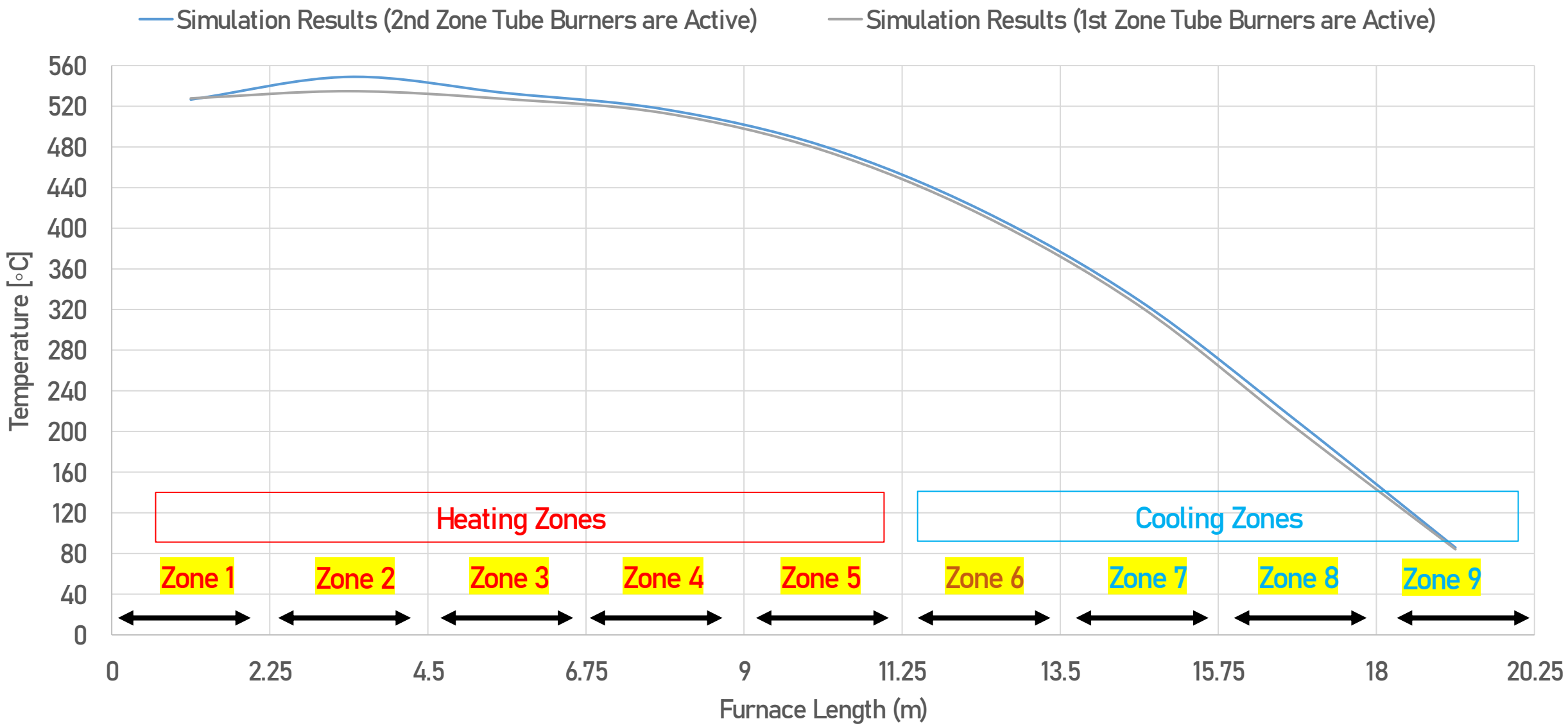
	Fixed Wall Temperature	Inlet Temperature and Velocity
Active Heaters	677 ◦C (950 K)	4 m/s, 677 ◦C
Passive Heaters	527 ◦C (800 K)	-

color	approximate temperature		
	°F	°C	K
faint red	930	500	770
blood red	1075	580	855
dark cherry	1175	635	910
medium cherry	1275	0690	0965

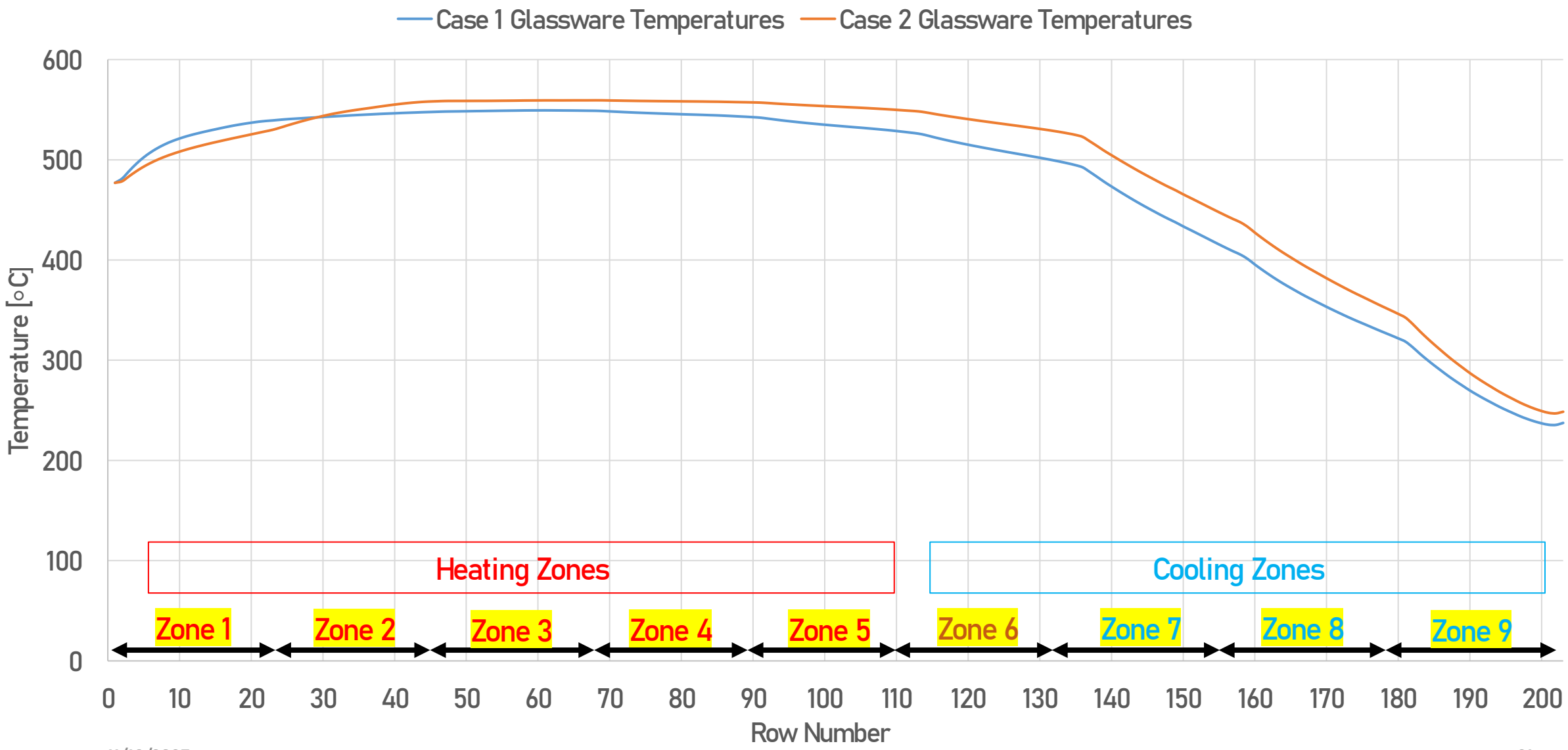
[2] Glow Colors by Temperatures Chart of Mild Steel, Retrieved from: https://www.hearth.com/talk/wiki/know-temperature-when-metal-glows-red/

Typical Glass Annealing Furnace View from Inlet

❖ Zone Temperature Comparison : Case 1 vs. Case 2.

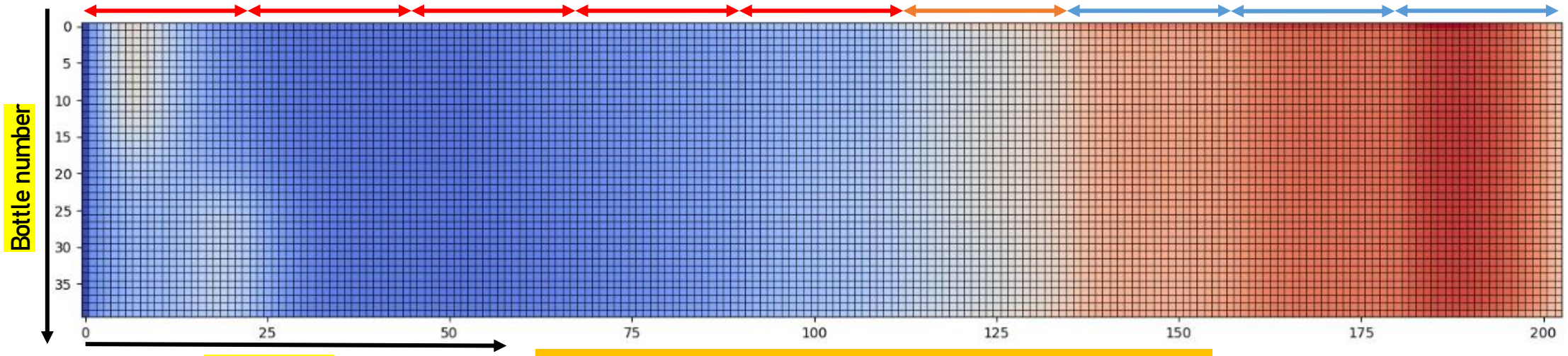


❖ Averaged Glassware Temperatures : Case 1 vs. Case 2

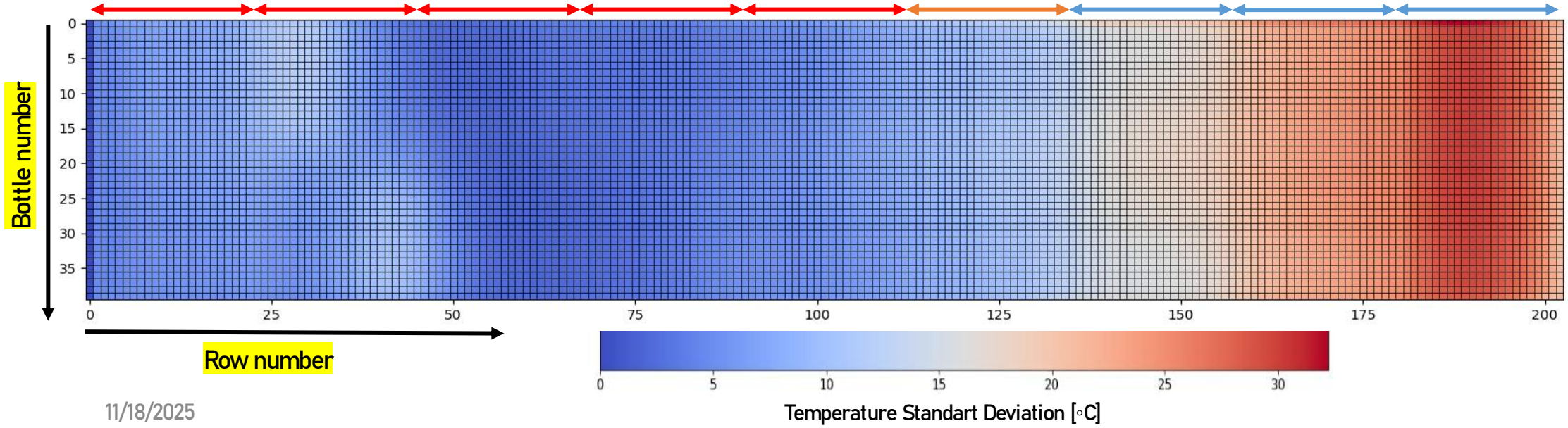


❖ Standart Deviation of Glass Temp. is used as a measure for Quality.

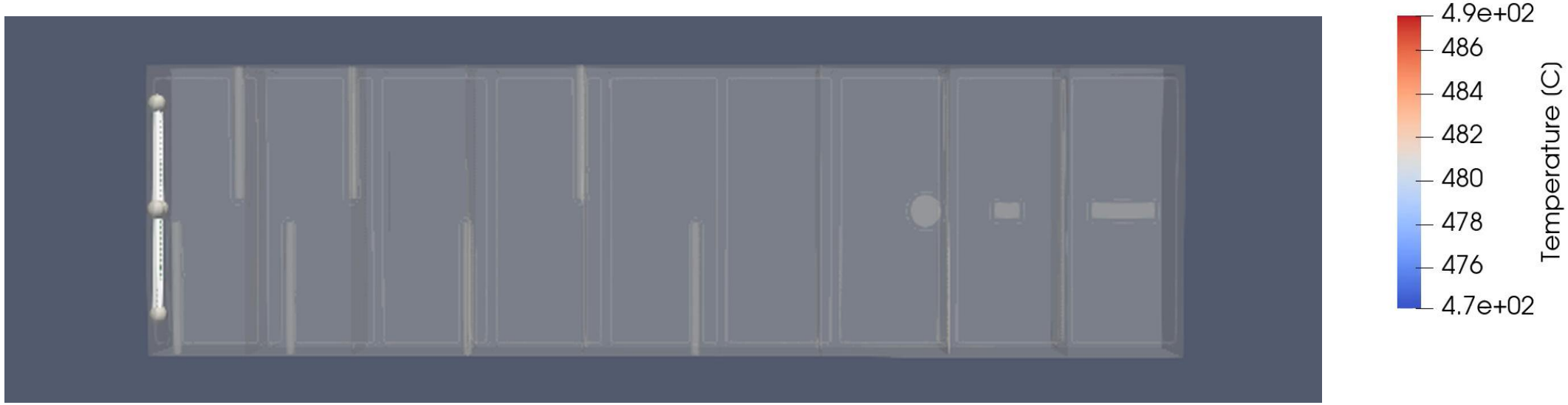
Case 1 : Active Tube Burners in Zone 1



Case 2 : Active Tube Burners in Zone 2



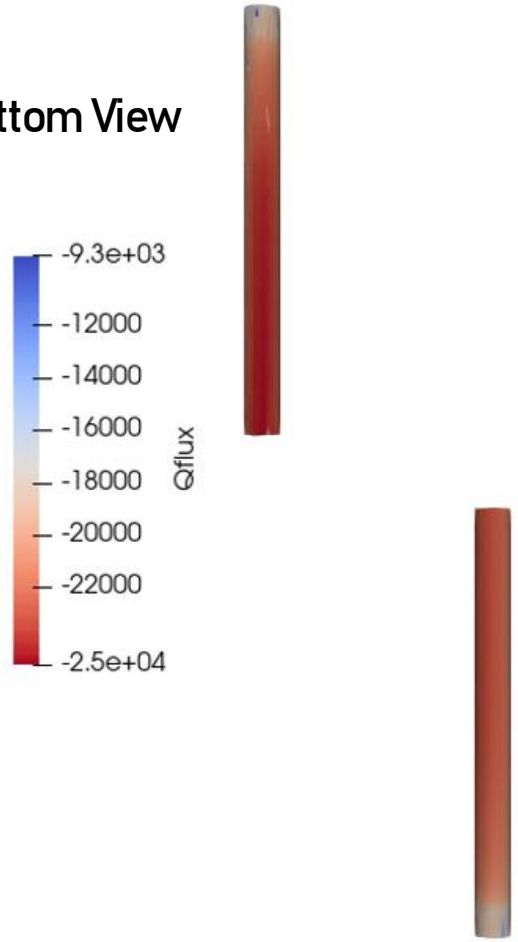
❖ Results: Glassware Temperatures Along the Annealing Furnace



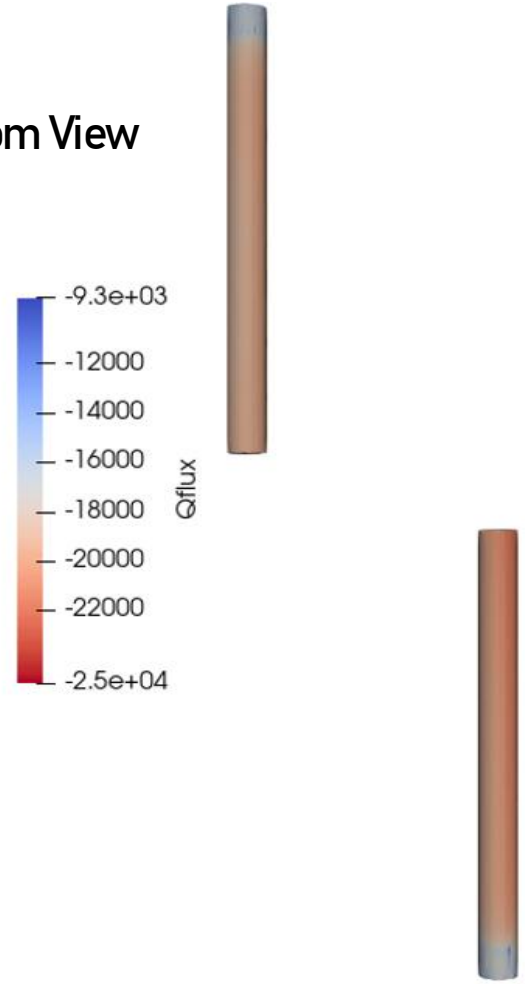
❖ Results: Radiative Flux Case 1 vs Case 2

- ❑ The Radiative heat fluxes of **Active Burners** for Case 1 and Case 2 are compared.
- ❑ Case 2 Results in Lower Heat Fluxes to the Glassware vs Case 1.

Case 1: Bottom View



Case 2: Bottom View



❖ Results: Net Radiative Heat Transfer Case 1 vs Case 2

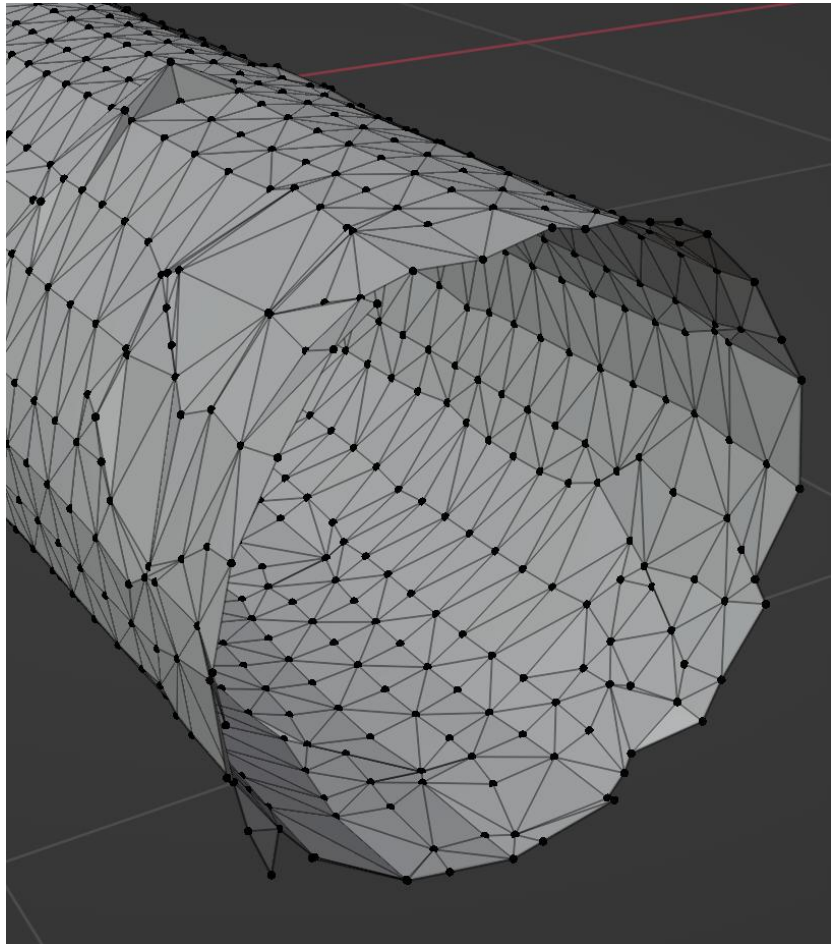
- To compare energy efficiency, Net Radiative Heat Transfers of the two cases are compared
- Because of the fixed wall temperatures, Passive Heaters (except Zone 5), has net heat Input rather then a net Output
- The following are the results for each zone for each case:

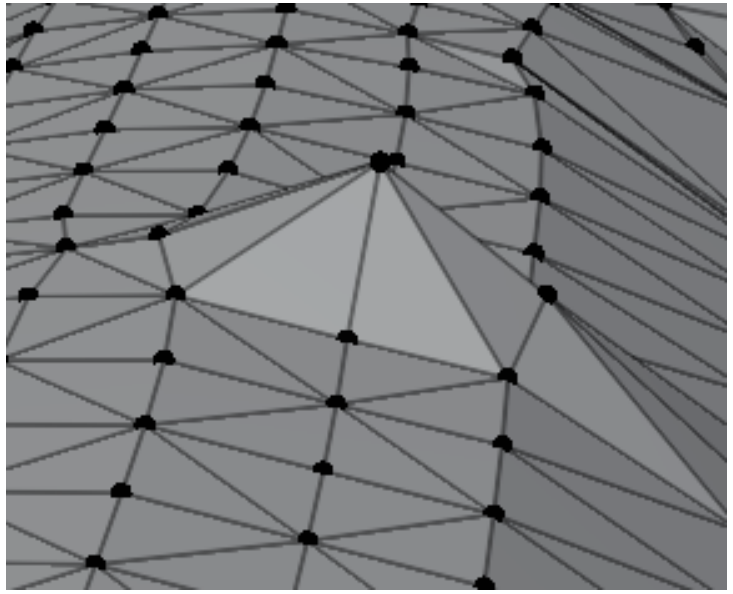
Zone Number	Case 1	Case 2
Zone 1	-61.4 kW	0.61 kW
Zone 2	0.74 kW	-55 kW
Zone 3	0.35 kW	0.38 kW
Zone 4	0.24 kW	0.26 kW
Zone 5	-0.54 kW	-0.47 kW

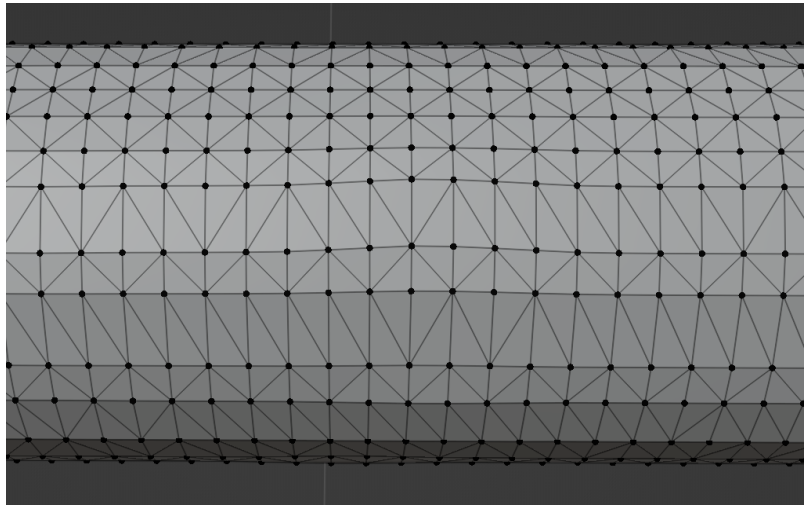
❖ Summary & Conclusions

- A thermal model of an industrial scale continuous annealing furnace was developed and validated using zone temperature data.
- Effect of position of active tube burners were studied :
  - **Case 1**: Active tube burners are in Zone 1
  - **Case 2**: Active tube burners are in Zone 2
- Case 2 results in higher glassware temperatures throughout the furnace compared with Case 1, except at Zone 1.
- Case 2 has more uniform temperature distribution of the glassware in the heating zones compared to Case 1
- A shift in the cooling zone is observed with Case 2 however the glassware temperatures at the outlet is similar
- Comparing the Radiative Heat Fluxes of the **active burners**, Case 2 has lower Radiative Heat Fluxes while maintaining higher temperatures → Increased Energy Efficiency
- Comparing Net Radiative Heat Transfers of the Active Heaters, Case 2 has lower Output than Case 1, → Less energy to maintain the wall temperature → Increased Energy efficiency

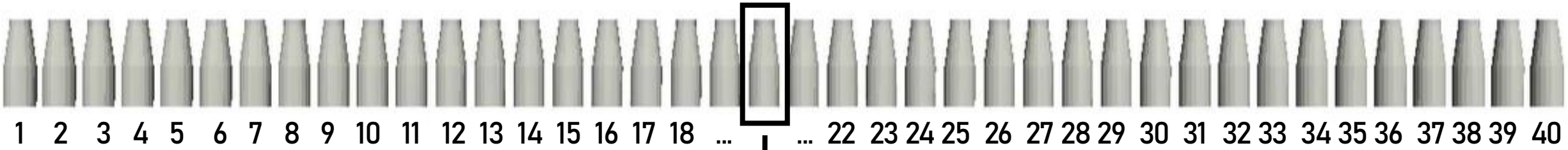
Backup







BACKUP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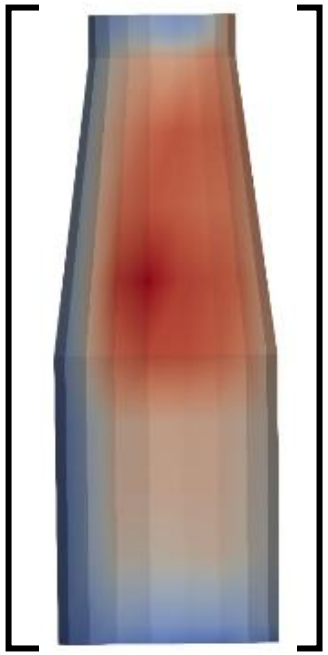
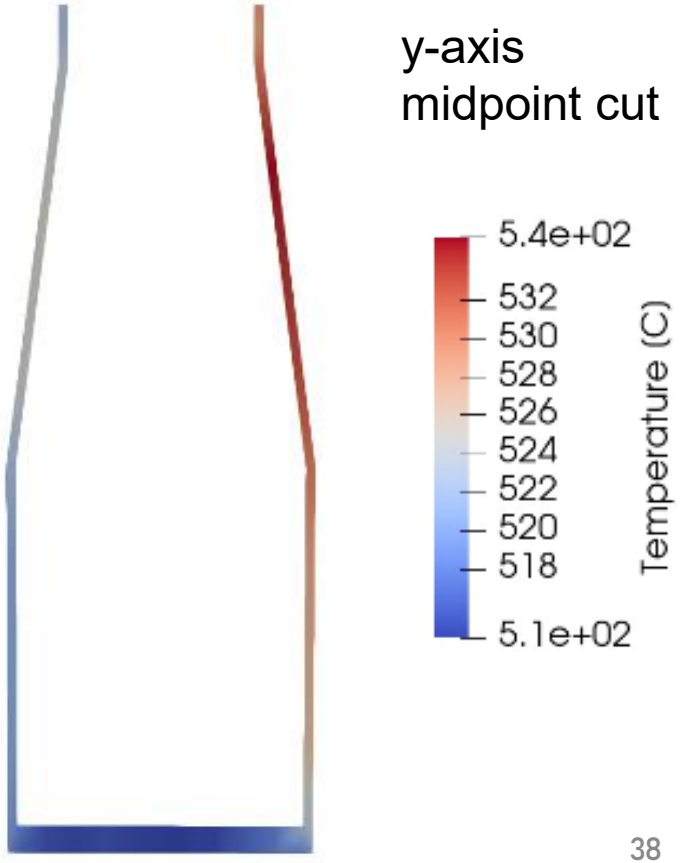
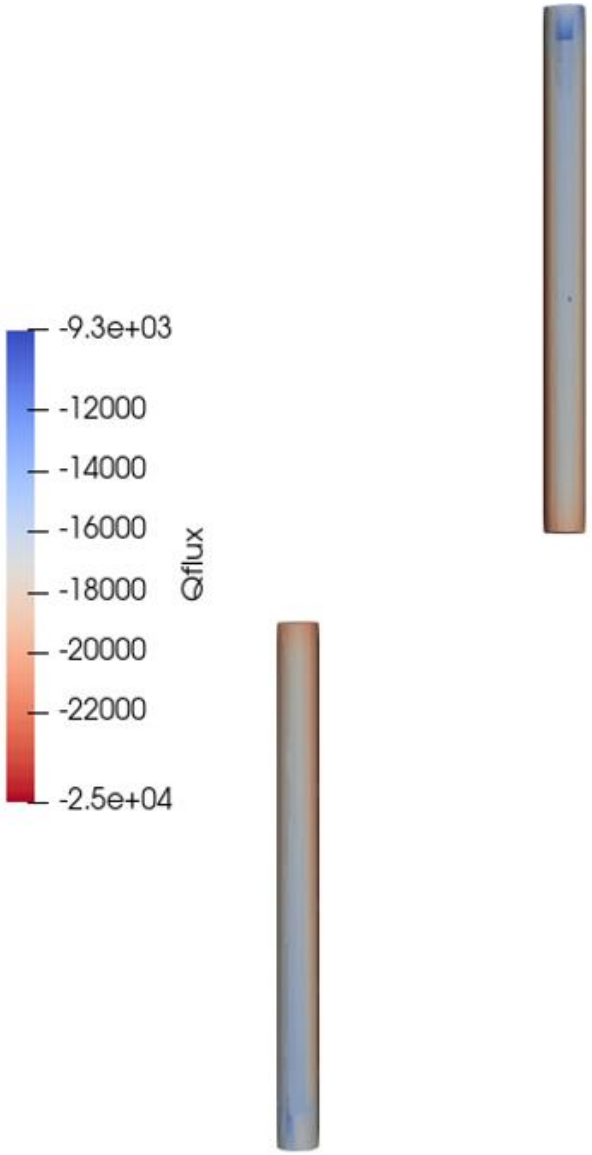
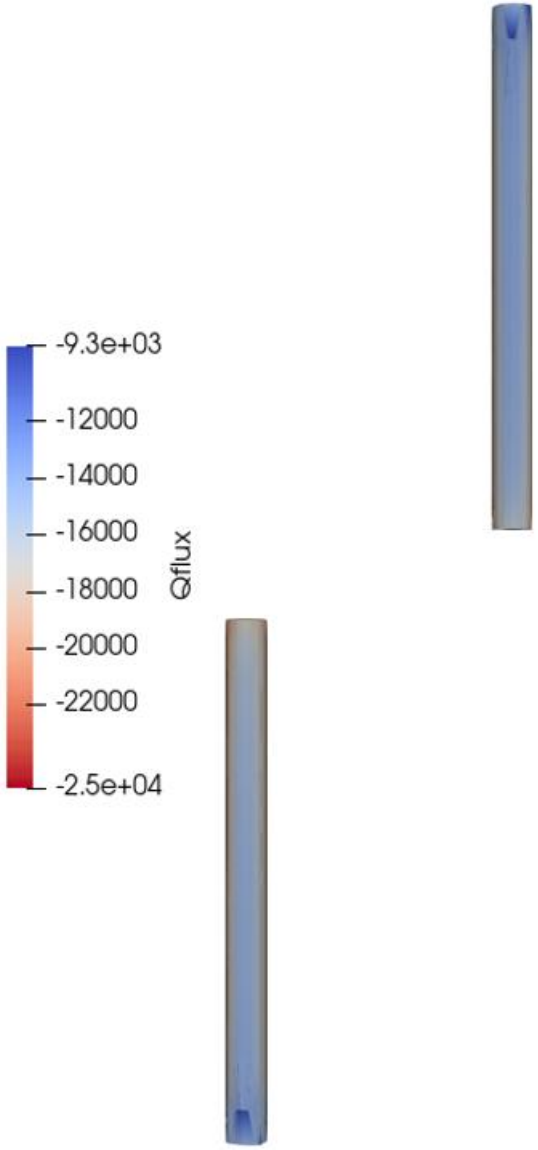



y-axis
midpoint cut
5.4e+02
532
530
528
526
524
522
520
518
5.1e+02
Temperature (C)

Backup

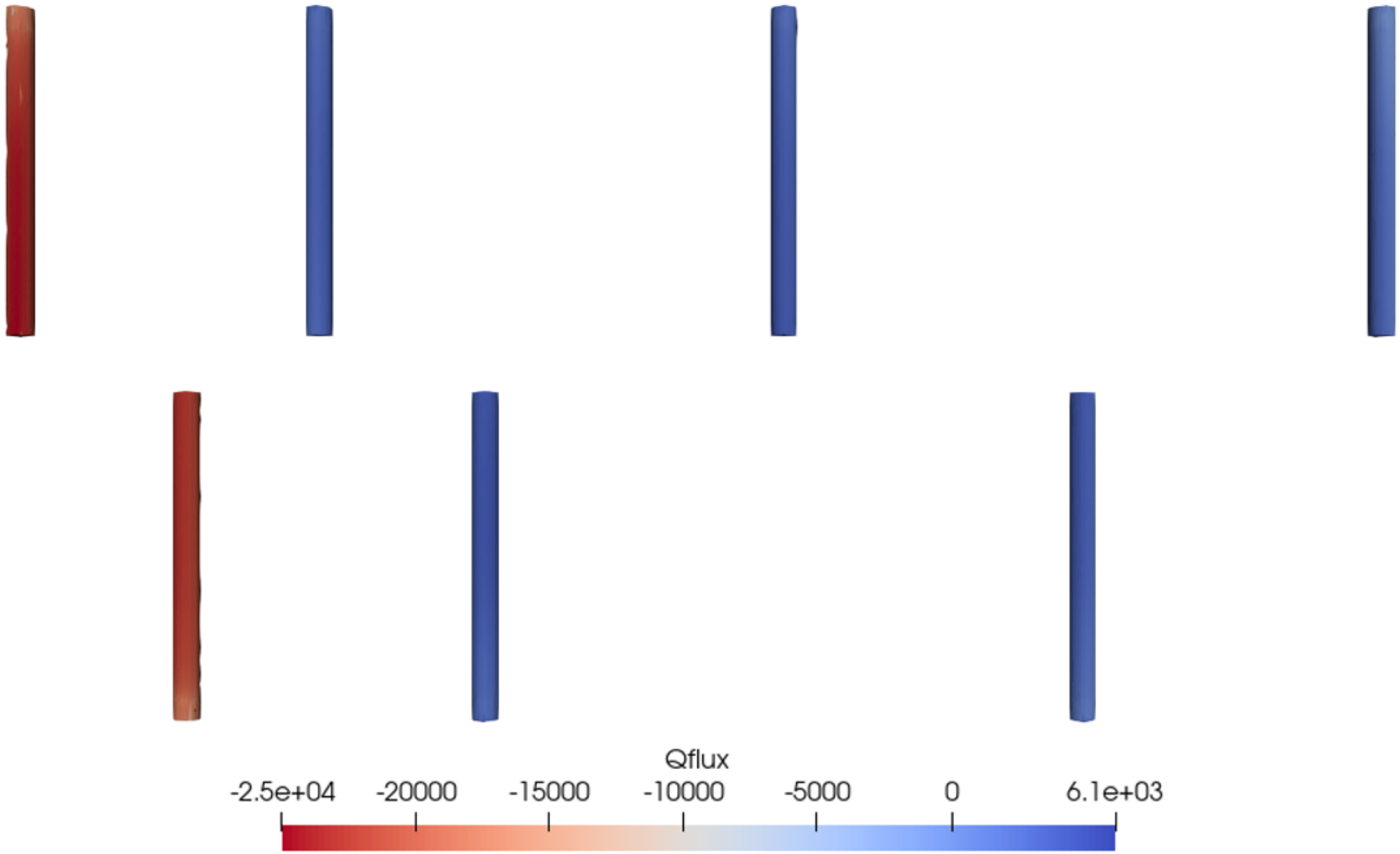
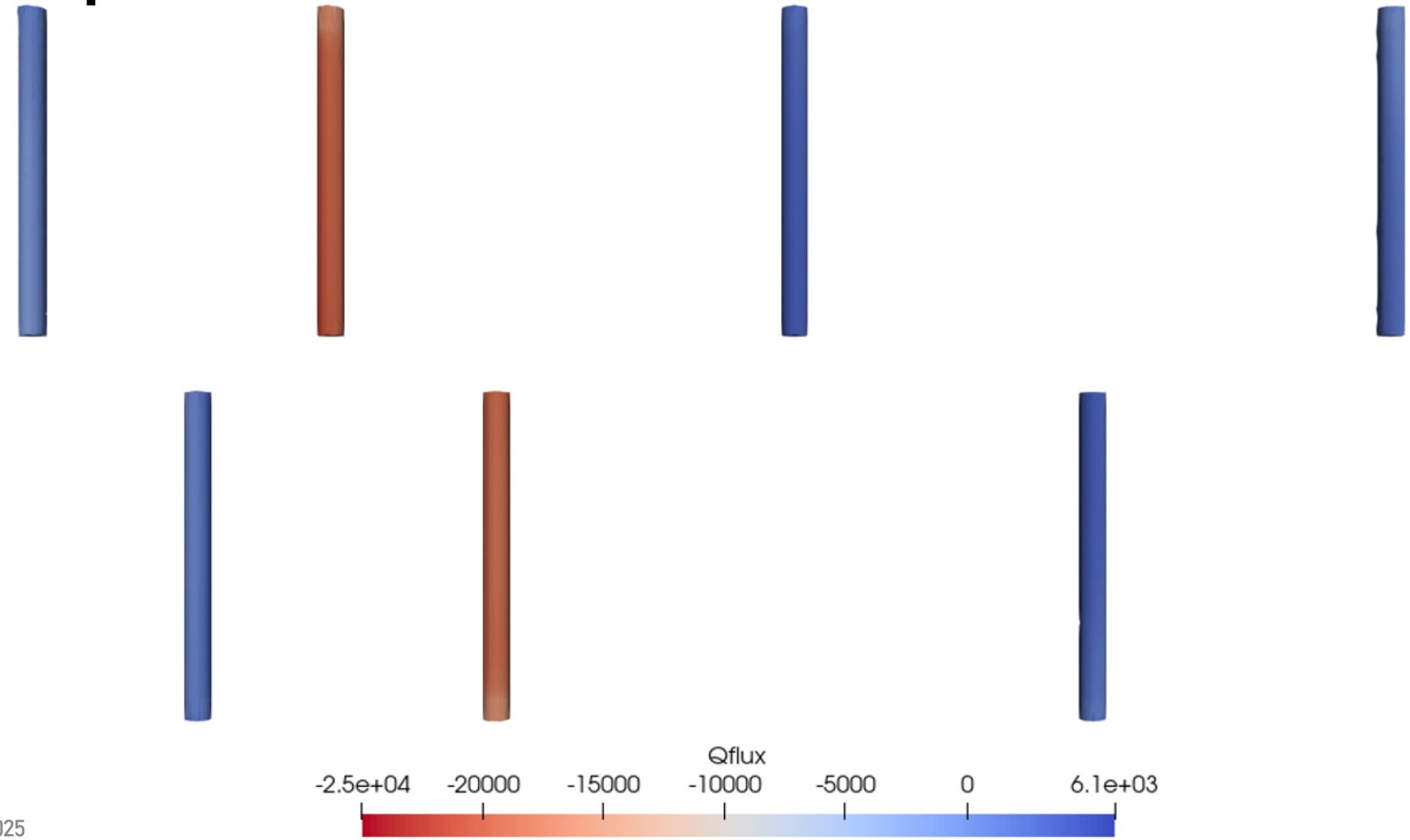


Case 1 Top View



Case 2 Top View

Backup



Backup